# A Framework to Model Stellar Irradiated Disks

## with Frequency-dependent Absorption and Scattering Opacities in Athena++

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3rd PFITS+ Meeting on Meeting on Hydrodynamics of Circumstellar Disks and Planet Formation
SETI Institute
June 12, 2026

# Context

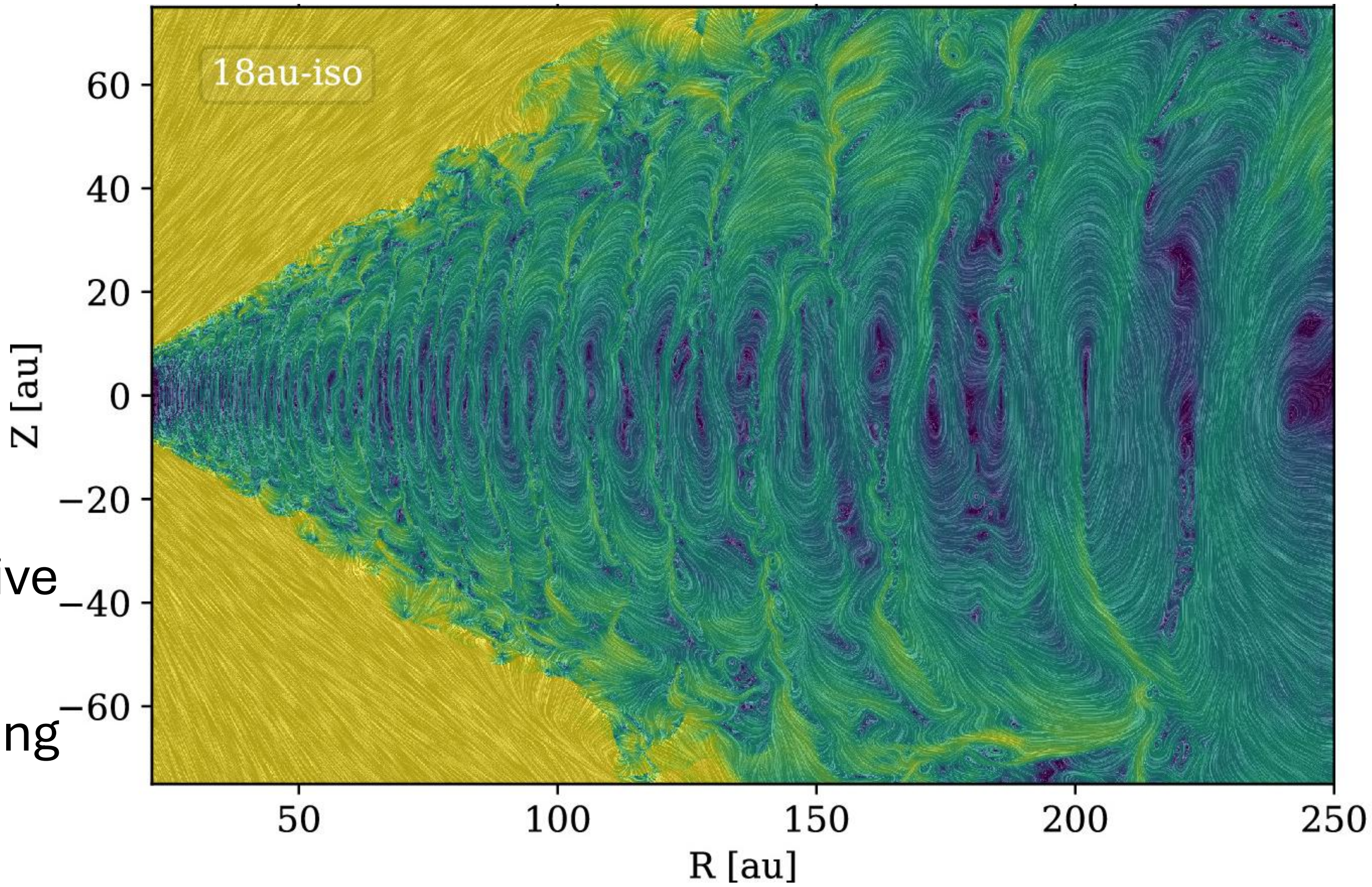
# Hydrodynamic Instabilities

Driven by

- Self-gravity
- Baroclinicity
- Buoyancy

Thermodynamically Sensitive

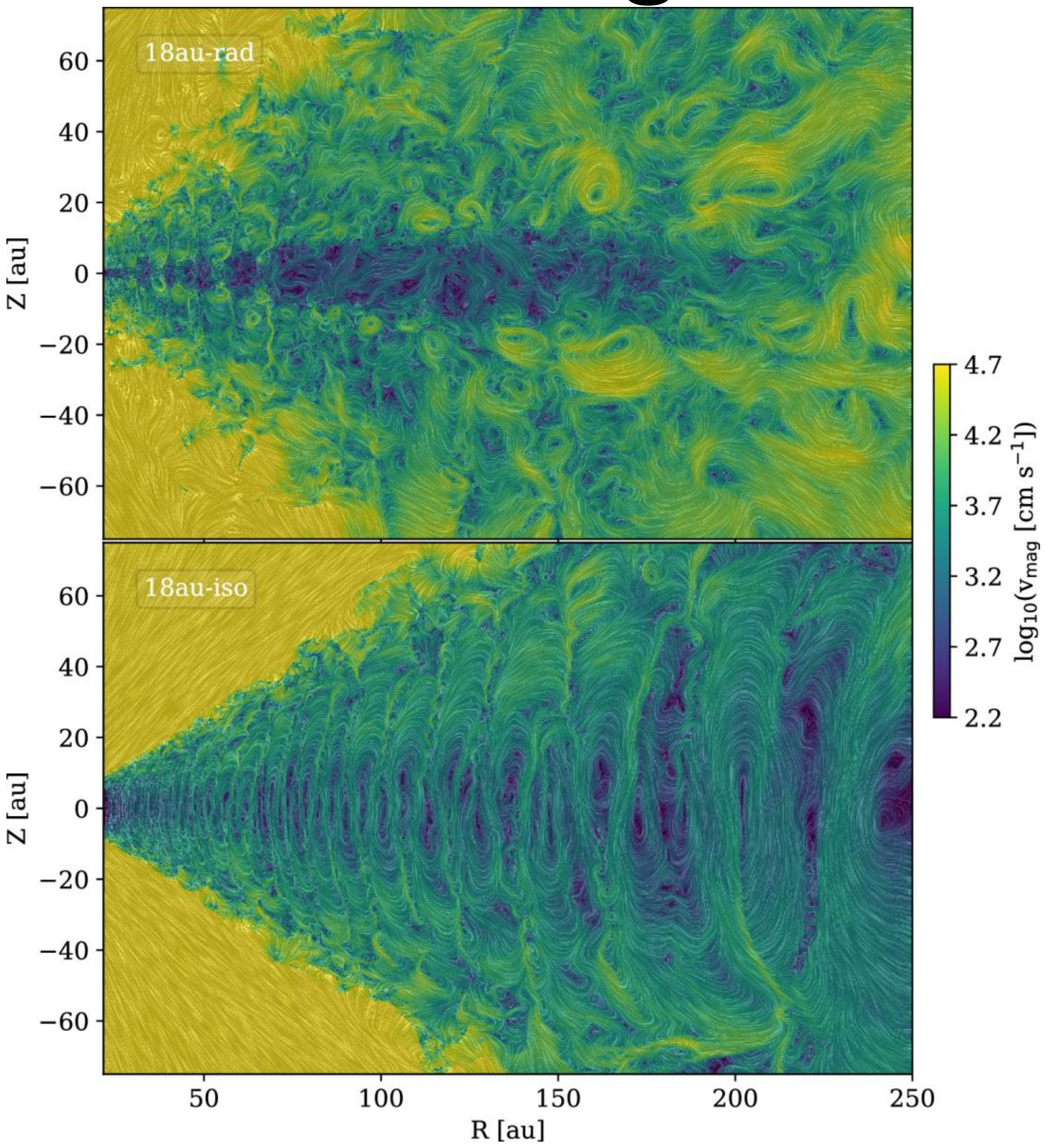
- (Classical) VSI
  - Requires instant cooling




Zhang+ 24

# Radiation-Hydrodynamics of Zhang+ 24

- Quiescent midplane
  - More conducive to planetesimal formation
- Turbulent surface layers
  - $v_{\mathrm{mag}}$ consistent with observed non-thermal line broadening?
- Caveats
  - Frequency-independent (gray)
  - Ray-traced irradiation
    - No radiation pressure
    - No scattering effects

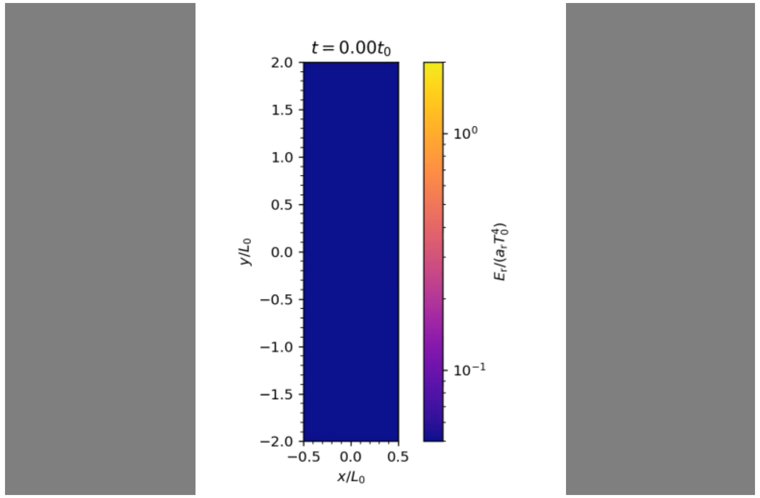


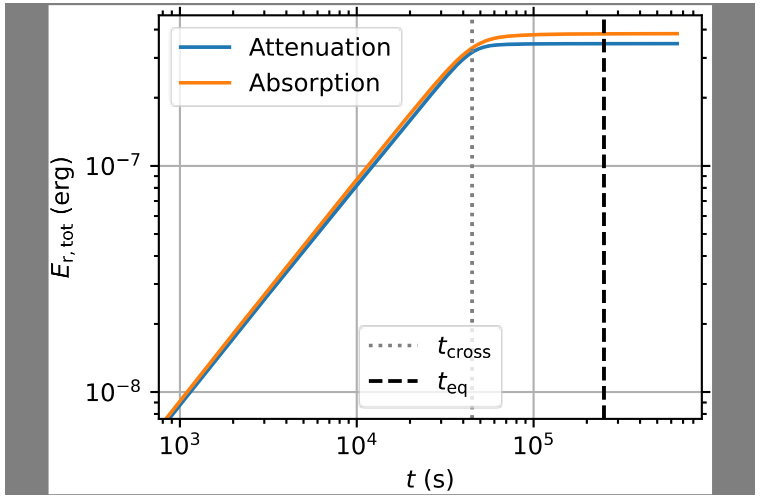

Zhang+ 24

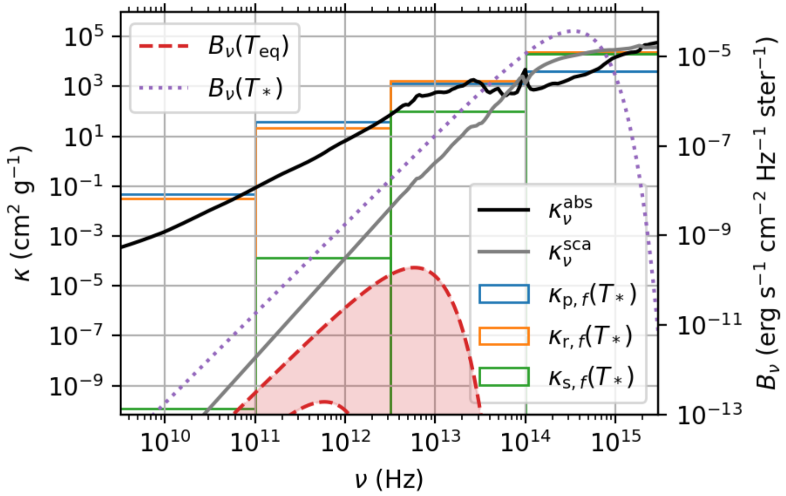
# Methodology

# Athena++ Multigroup Radiation Transport

Jiang 21




## Discrete Ordinates

**Avoids** moment-method limitations (e.g., intermediate optical depths, shadow casting, & beam crossing)

## Nonrelativistic

**Implicitly** solves *time-dependent* radiative transfer equation to avoid timestep limited by speed of light

## Frequency Dependence

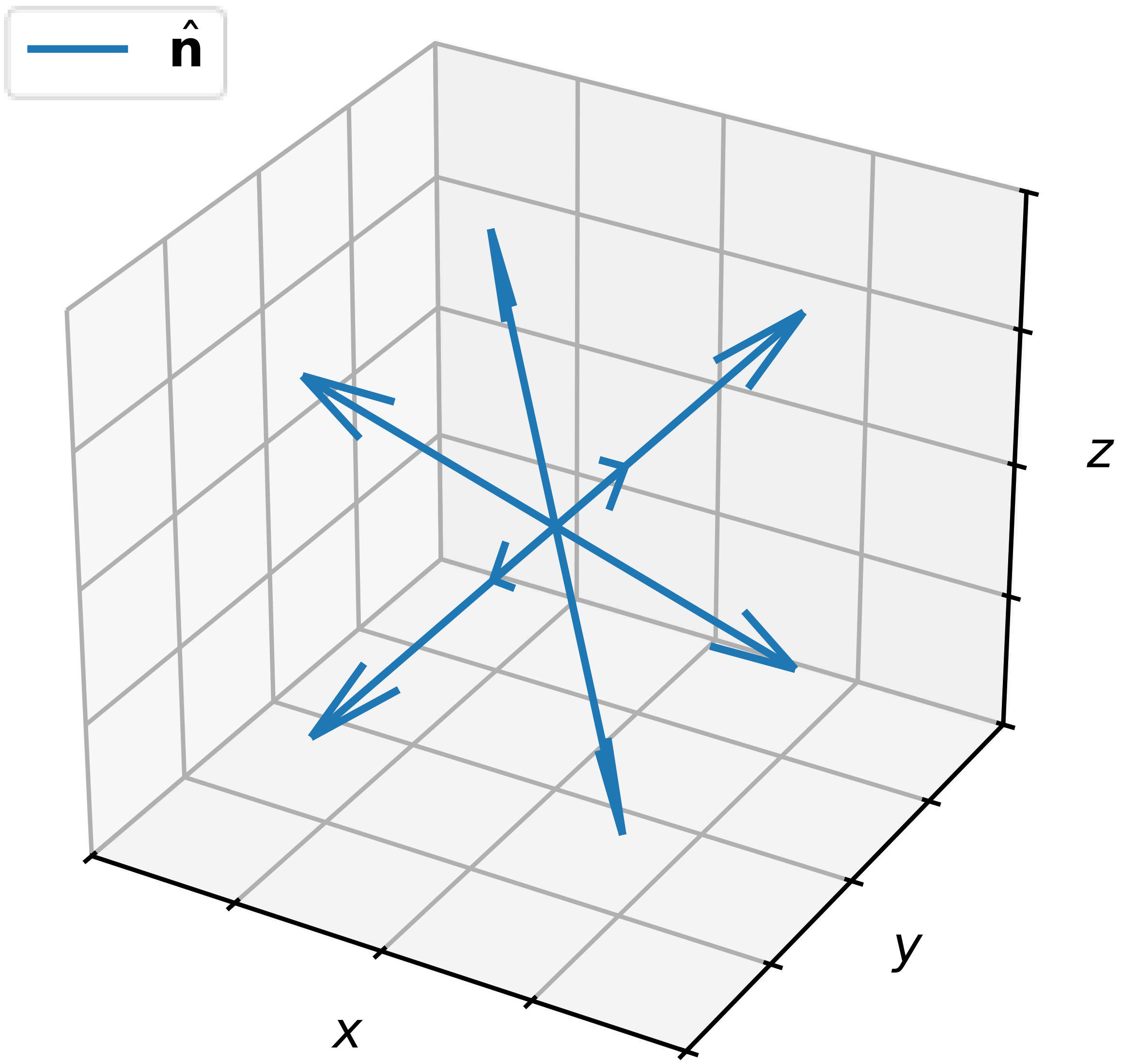
Supports any number of **custom-spaced groups** for band-mean absorption and **scattering** opacities

***See Prakruti Sudarshan's presentation next***

# Angular Discretization

$$\frac{\partial I_f}{\partial t} + c\hat{\mathbf{n}} \cdot \nabla I_f = c\left(\overbrace{\eta_f}^{\text{emissivity}} - \overbrace{\chi_f I_f}^{\text{opacity}}\right)$$

solved in the lab frame along local **discrete ordinates** $\hat{\mathbf{n}}$.



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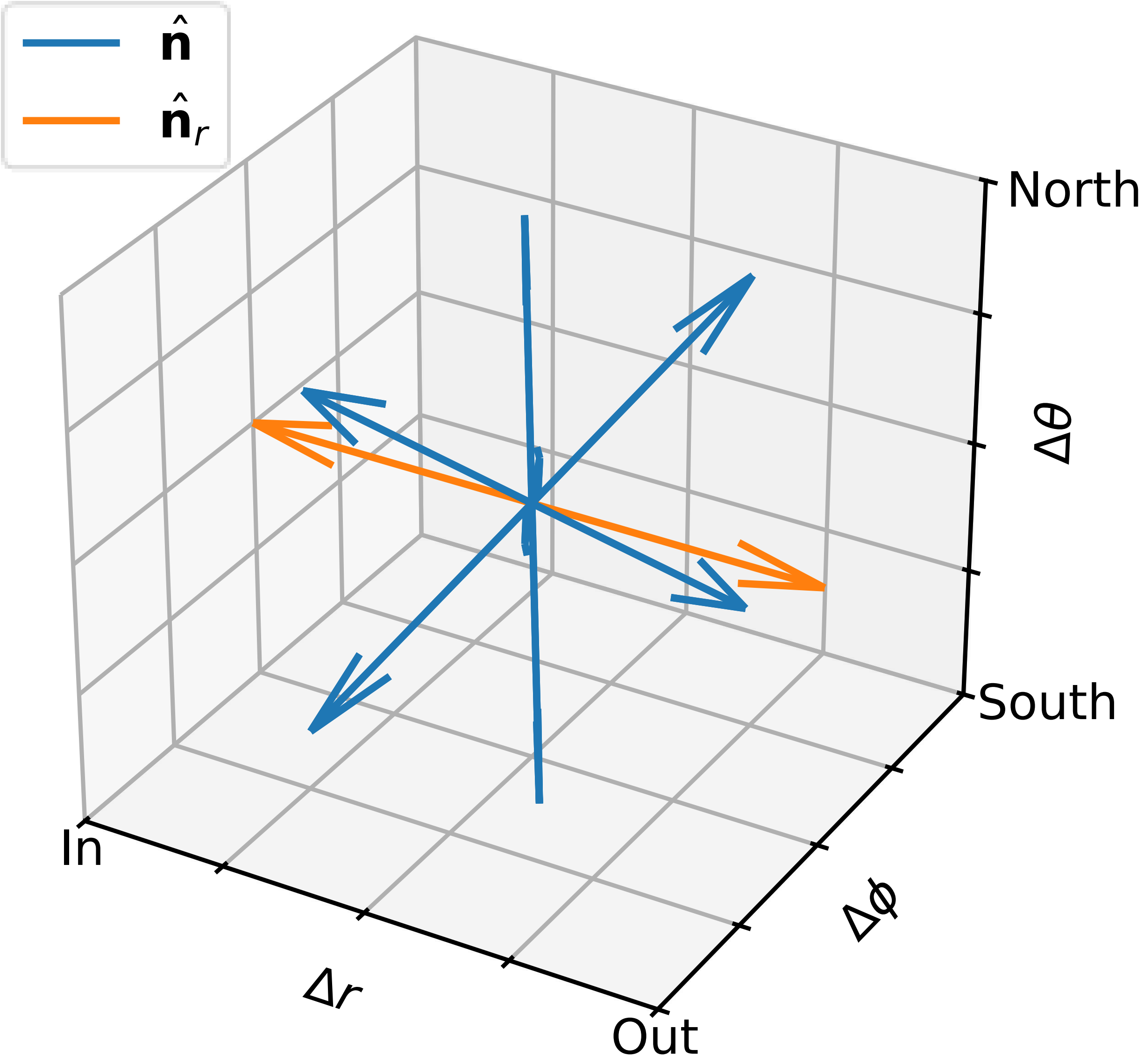
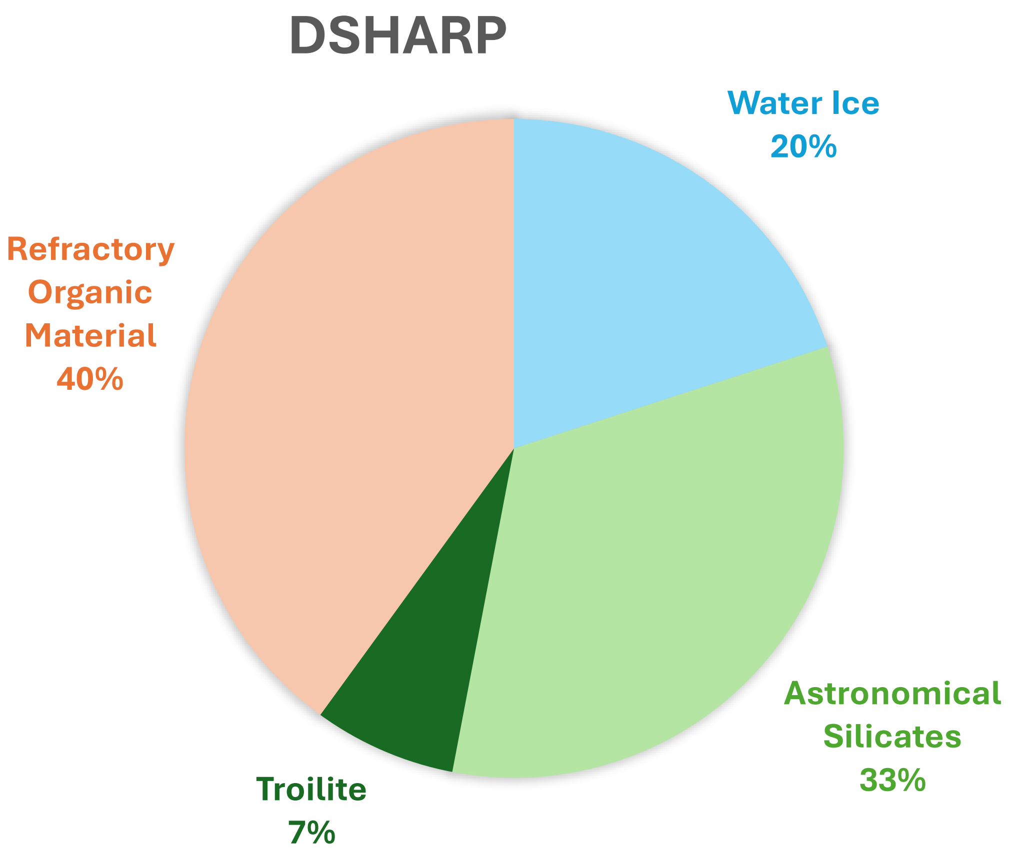
solved in the lab frame along local **discrete ordinates** $\hat{\mathbf{n}}$.

Radial rays for stellar irradiation in spherical coordinates



# Opacities

# Dust Composition




**Mie calculations:** $\kappa_\nu^{\rm abs}(a)$, $\kappa_\nu^{\rm sca}(a)$

Birnstiel+18

# Grain Size Distribution

$$n(a) \propto \begin{cases} a^{-q} & \text{for } a_{\min} \le a \le a_{\max}, \\ 0 & \text{otherwise} \end{cases}$$

$q = 3.5$ (MRN 77)

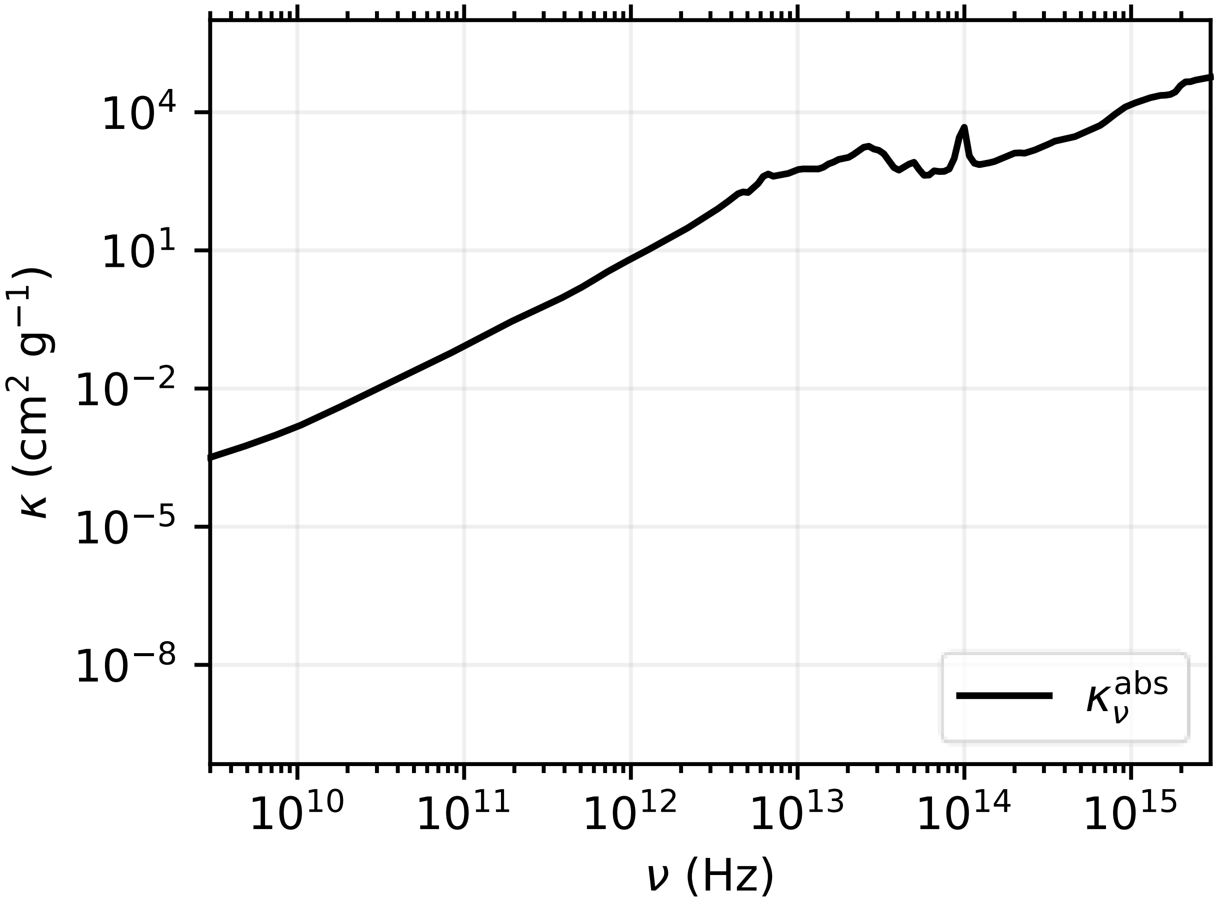
$a_{\min} = 0.1 \mu\text{m}$

$a_{\max} = 1 \mu\text{m}$

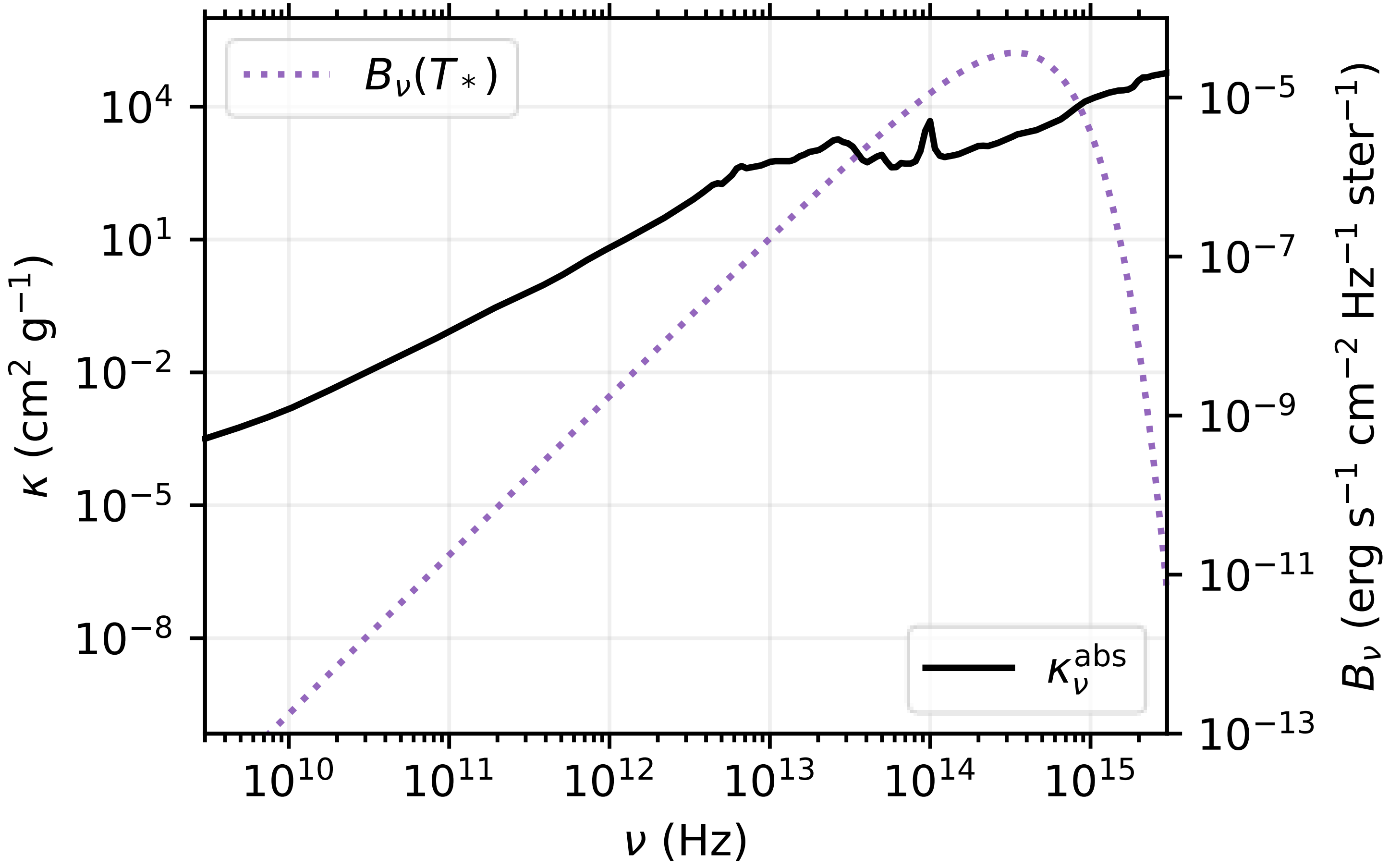
github.com/
birnstiel/
**dsharp_opac**

Birnstiel+18

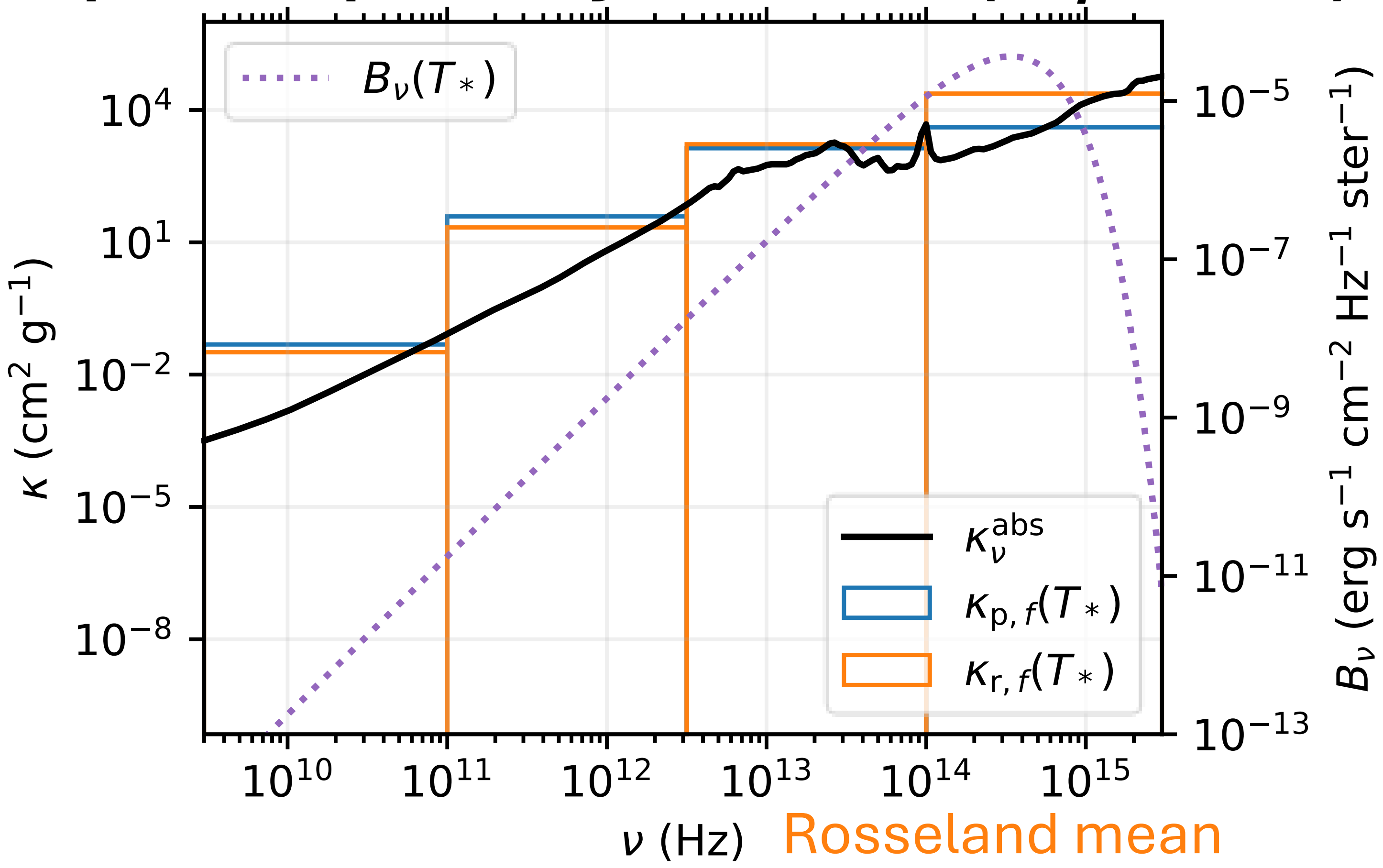
# Monochromatic Absorption Opacity



# Sunlike Blackbody Spectrum ($T_* = 5780$ K)



# Multigroup Frequency Bands ($N_f = 4$)



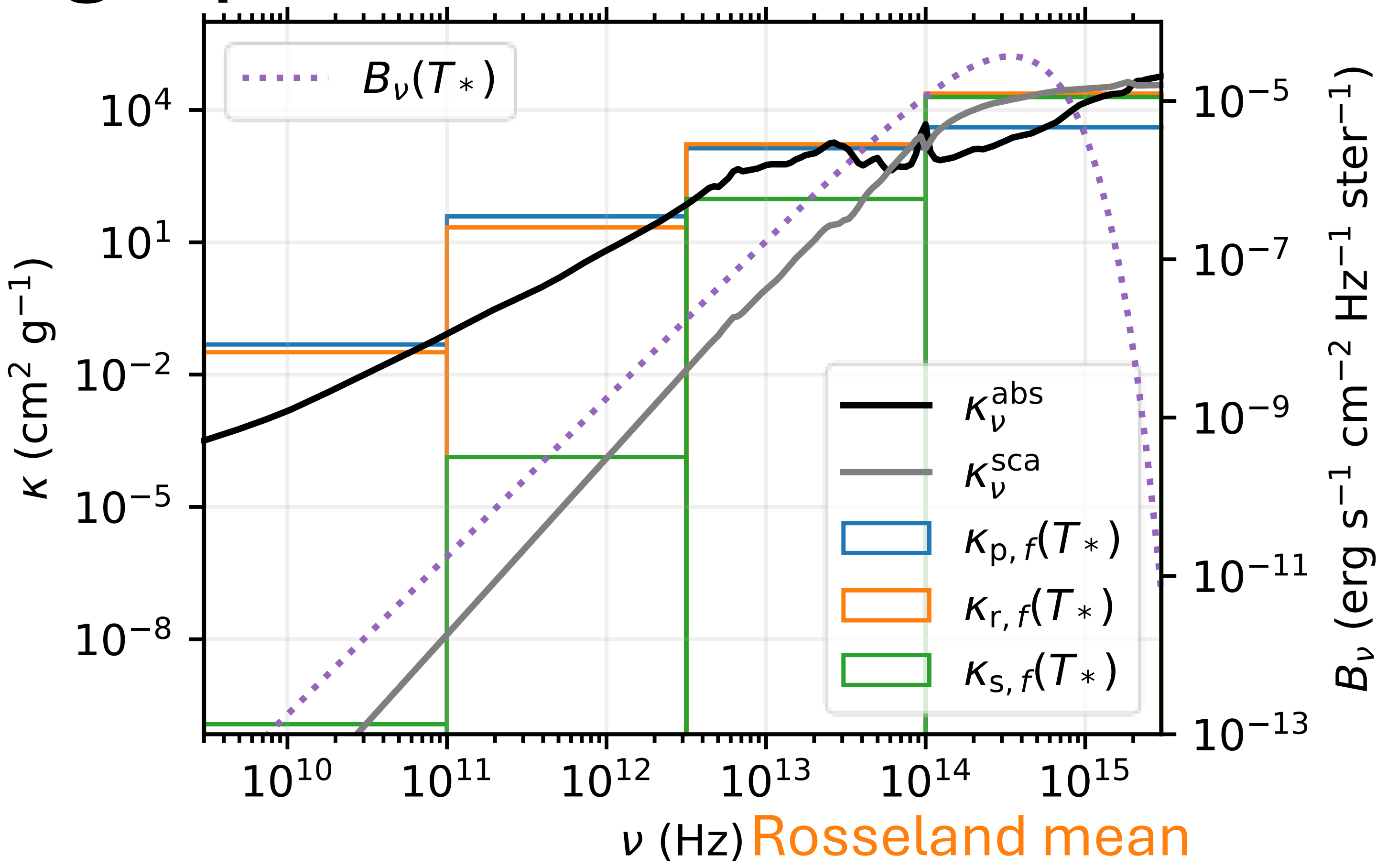

Planck mean

$$\kappa_{\mathrm{p},f}(T) \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \kappa_\nu^{\mathrm{abs}} B_\nu(T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} B_\nu(T) \mathrm{d}\nu}$$

Rosseland mean

$$\frac{1}{\kappa_{\mathrm{r},f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\mathrm{abs}}\right)^{-1} (\partial B_\nu / \partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu / \partial T) \mathrm{d}\nu}$$

# Scattering Opacities




Rosseland mean

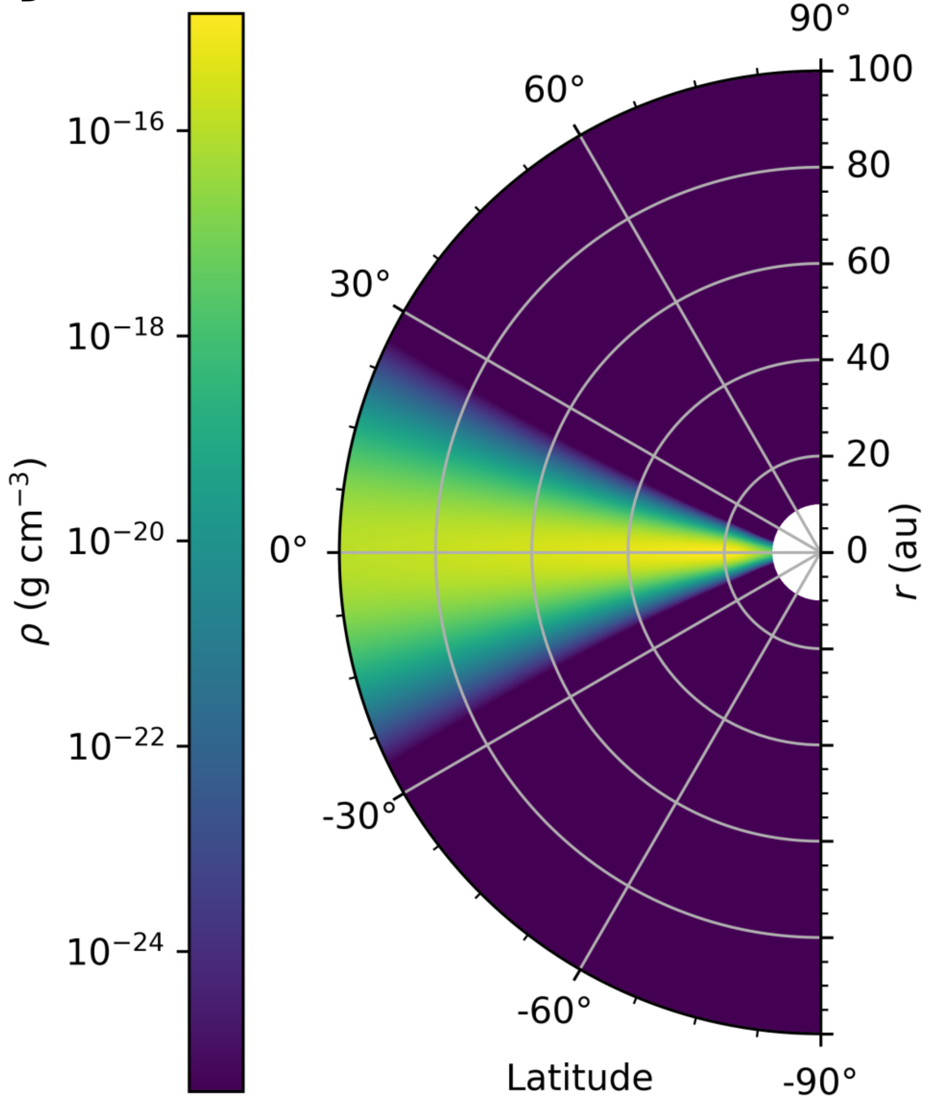
$$\frac{1}{\kappa_{s,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} (\kappa_\nu^{\mathrm{sca}})^{-1}\,(\partial B_\nu/\partial T)\mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu/\partial T)\mathrm{d}\nu}$$

$$\frac{1}{\kappa_{r,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\mathrm{abs}} + \kappa_\nu^{\mathrm{sca}}\right)^{-1}(\partial B_\nu/\partial T)\mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu/\partial T)\mathrm{d}\nu}$$

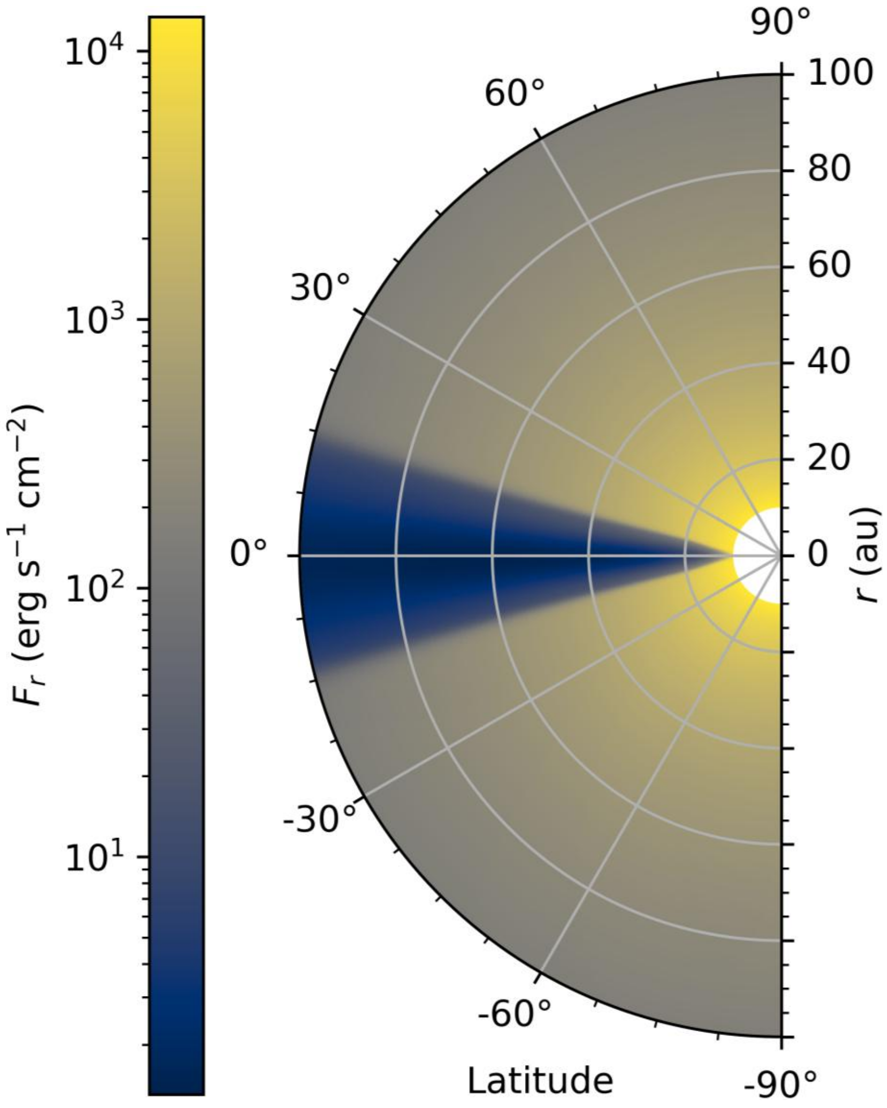
# Stellar-Irradiated Disk

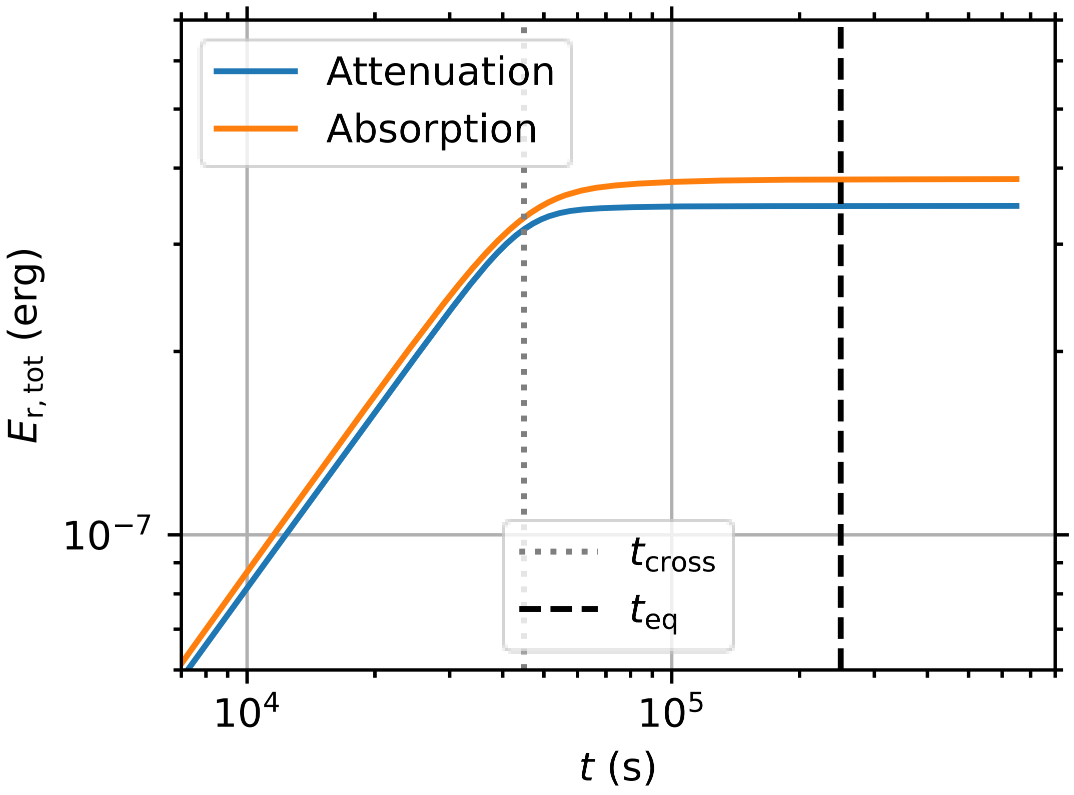
# Axisymmetric Model

$$\tau_*(r,\theta,\phi) = \int_{r_{\min}}^{r} \rho(r',\theta,\phi)\left(\kappa_*^{\mathrm{abs}} + \kappa_*^{\mathrm{sca}}\right)\mathrm{d}r'$$



# Hydrostatic Evolution



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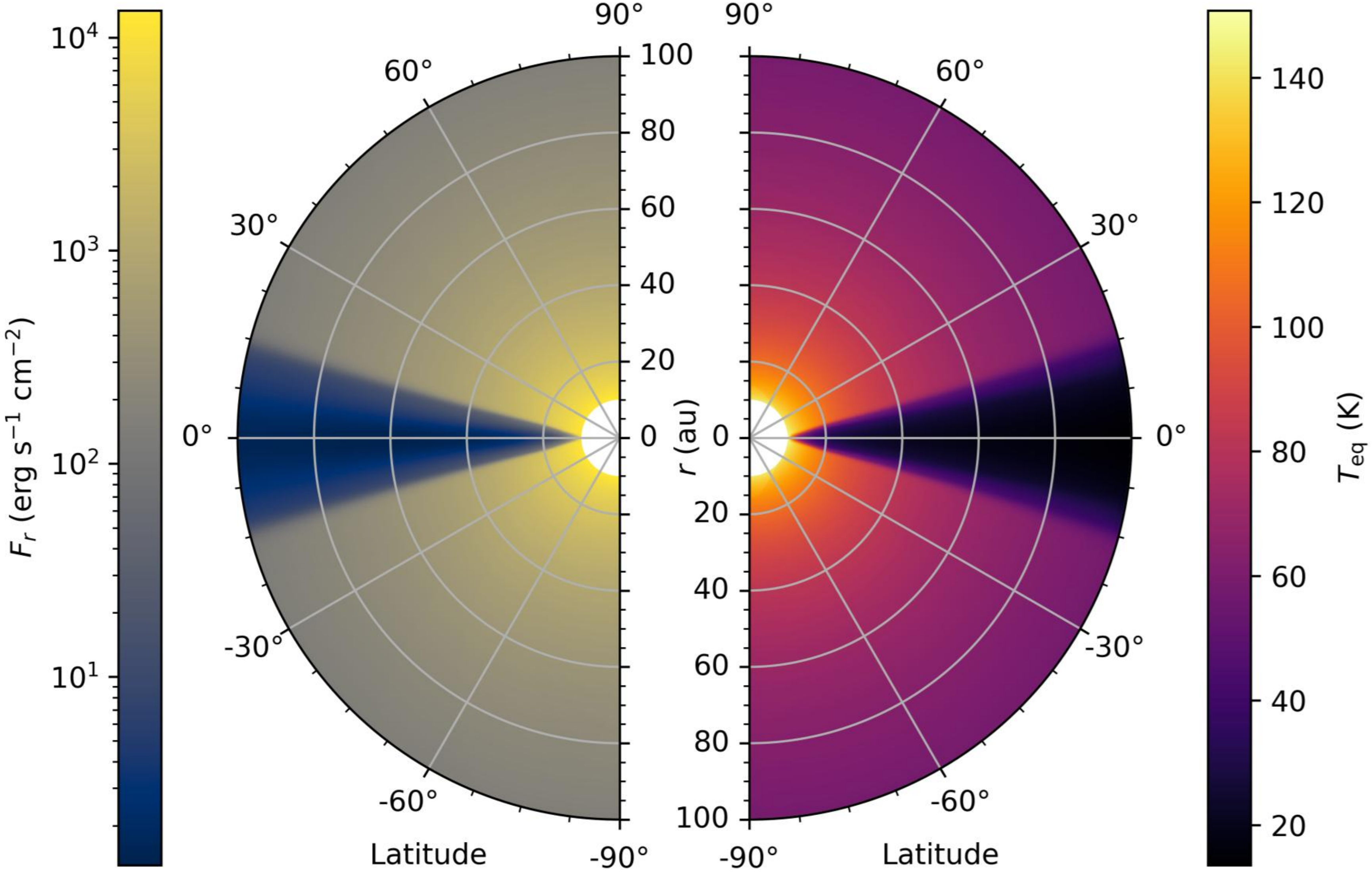
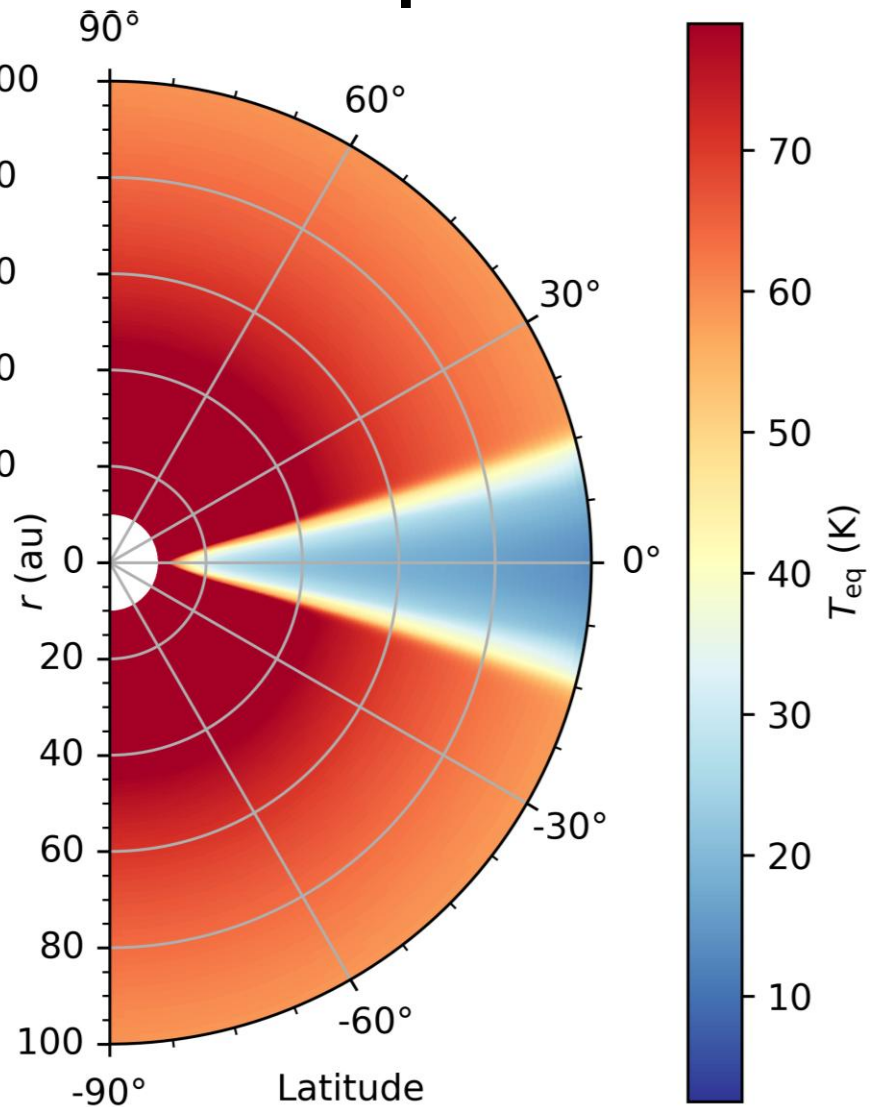
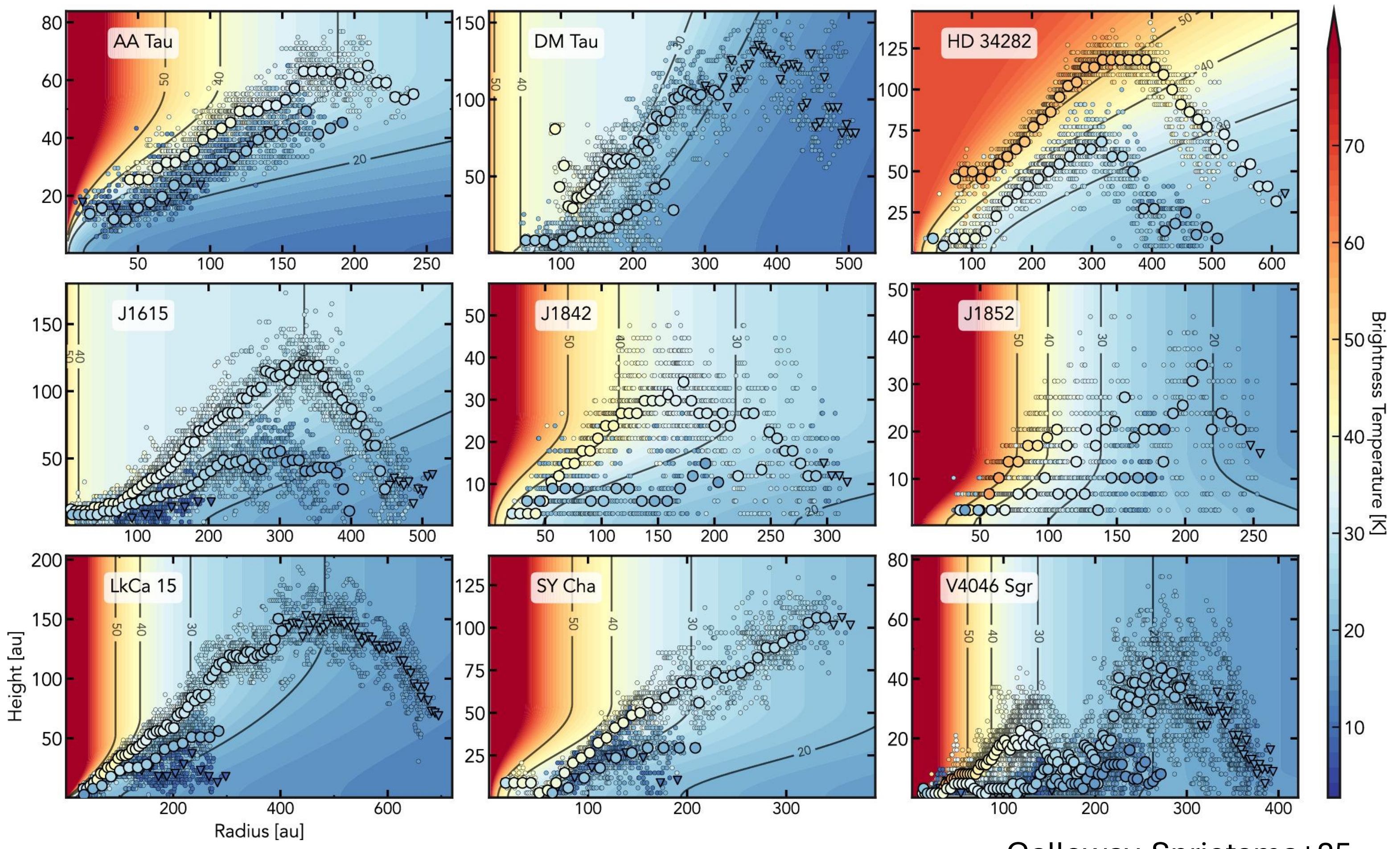


# Temperature Maps (exoALMA. V.)




Galloway-Sprietsma+25

# Hydrostatic Comparisons

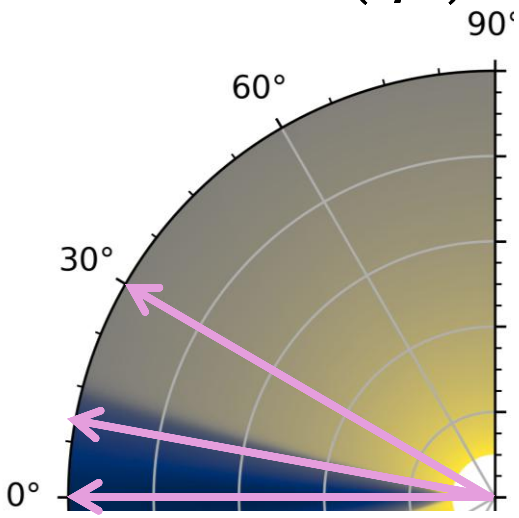
# Gray Opacity: **Without** Dust Emission

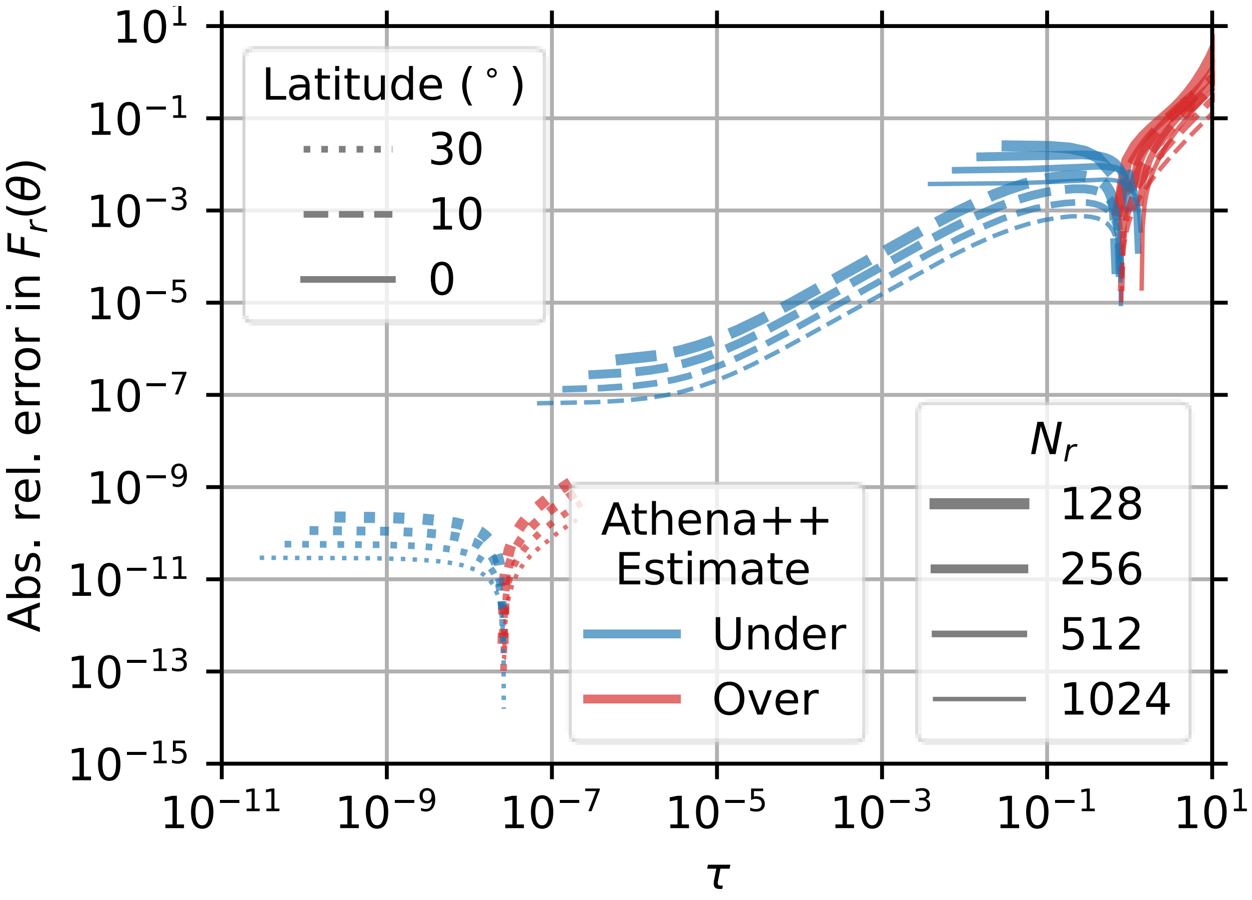
Formal radiative transfer solution

$$I(\tau) = I(0)e^{-\tau}$$

Analytic ray-tracing solution

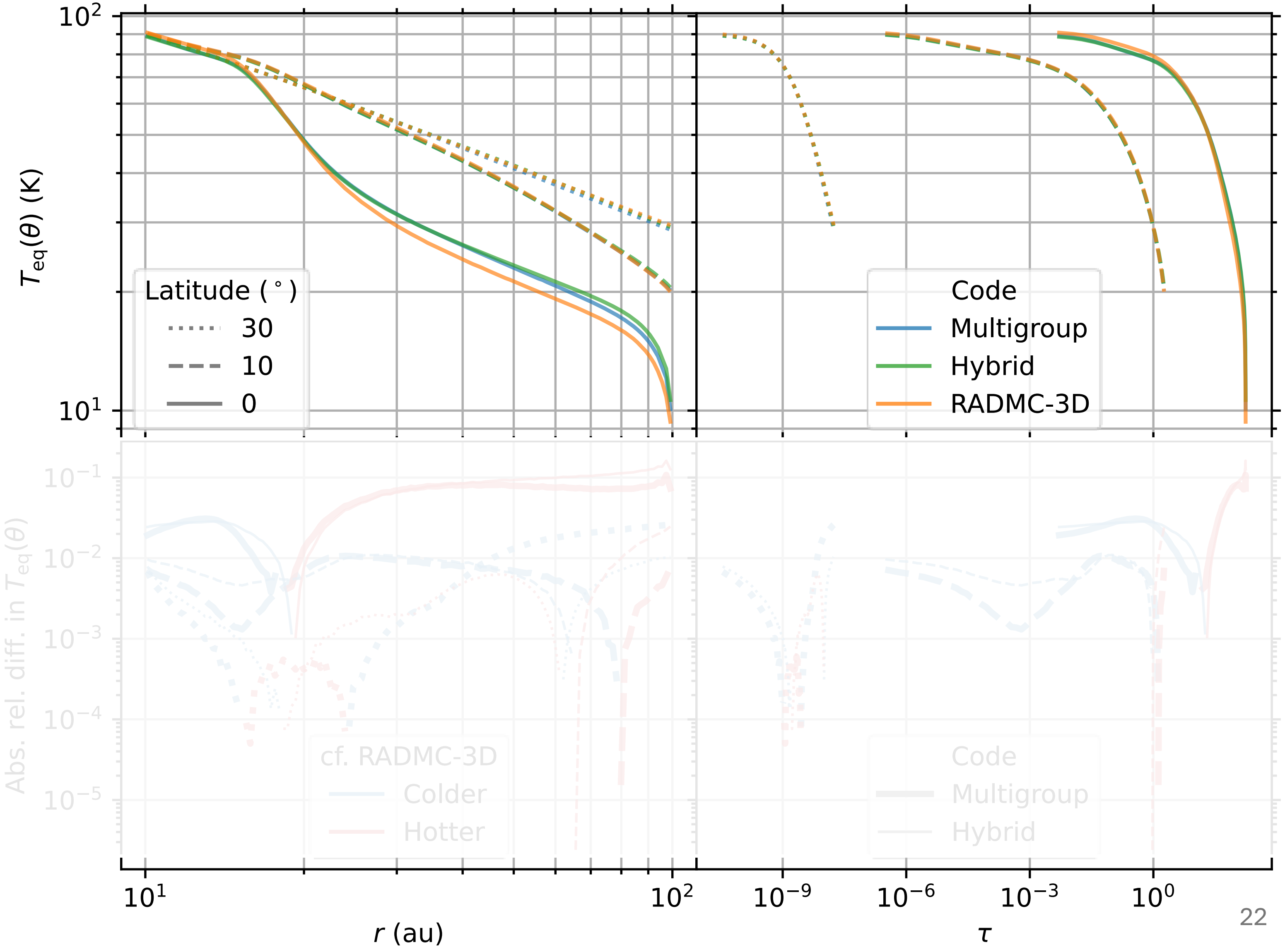
$$F_r(\tau) = \sigma T_*^4 \left(\frac{R_*}{r}\right)^2 e^{-\tau}$$



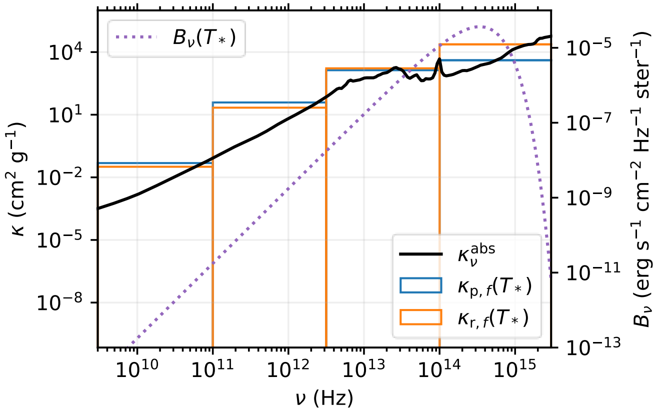
# Gray Opacity: **With** Dust Emission

- RADMC-3D
  - Monte Carlo radiative transfer
    - Standard benchmark
  - Static structure
- Hybrid (Athena++)
  - Zhang+24
  - Ray-traced stellar irradiation

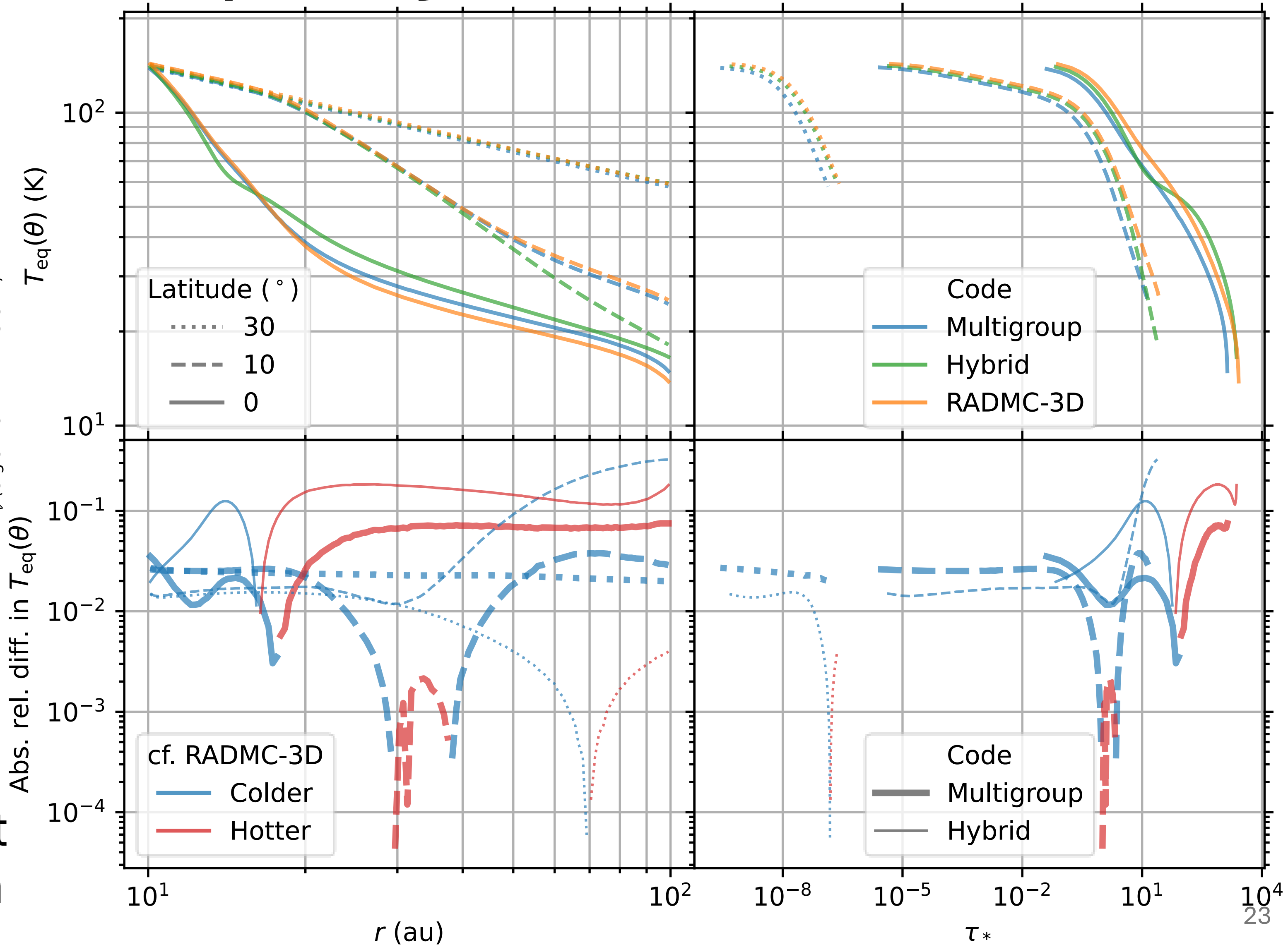


# Freq.-dependent Opacity: **With** Dust Emission

- 60%+ hotter atm.
- Colder midplane

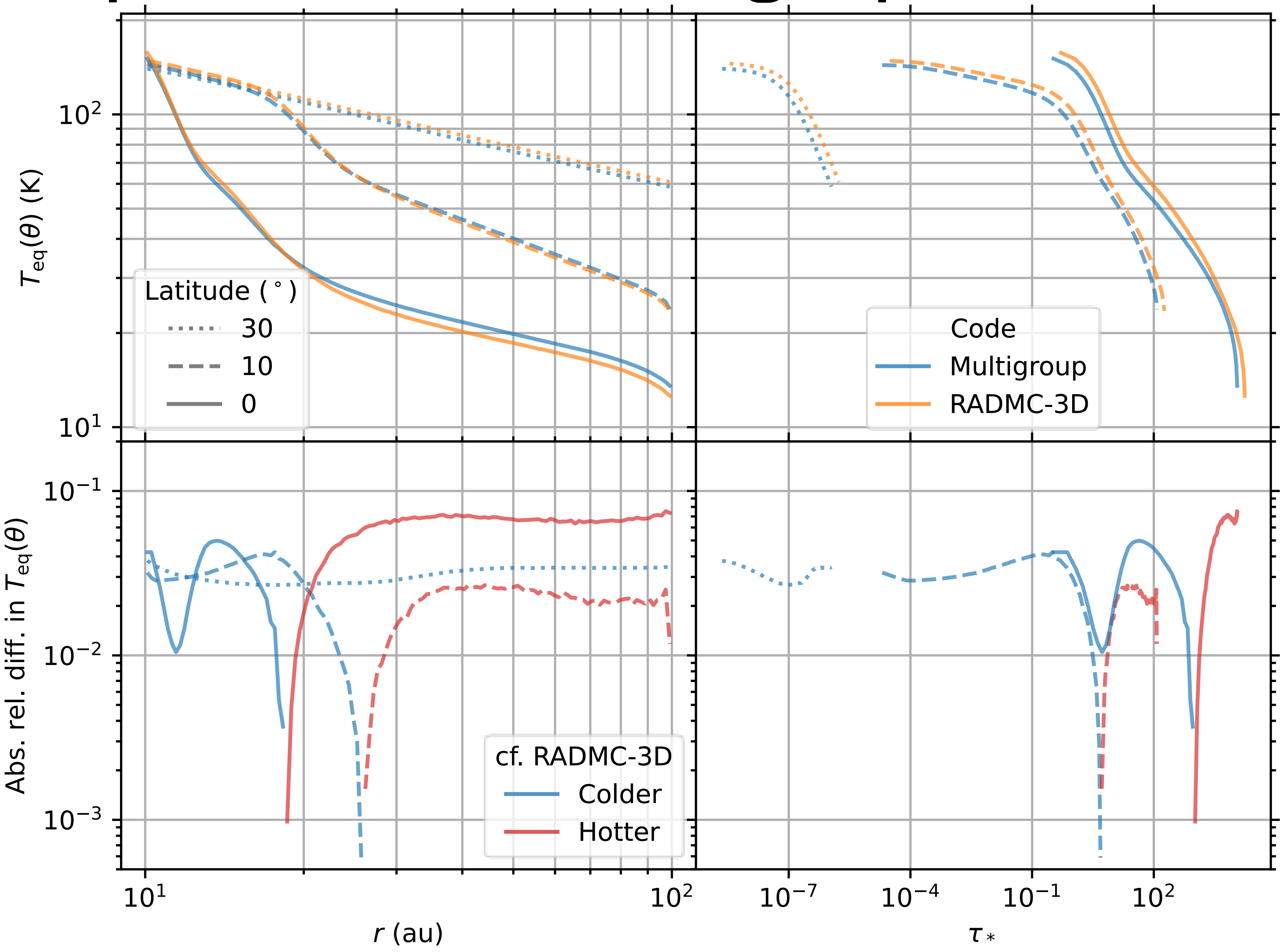



- Hybrid
  - Gray (dust) thermal emission
  - Rel. diff. greater at high optical depth



# Athena++ Absorption + Scattering Opacities

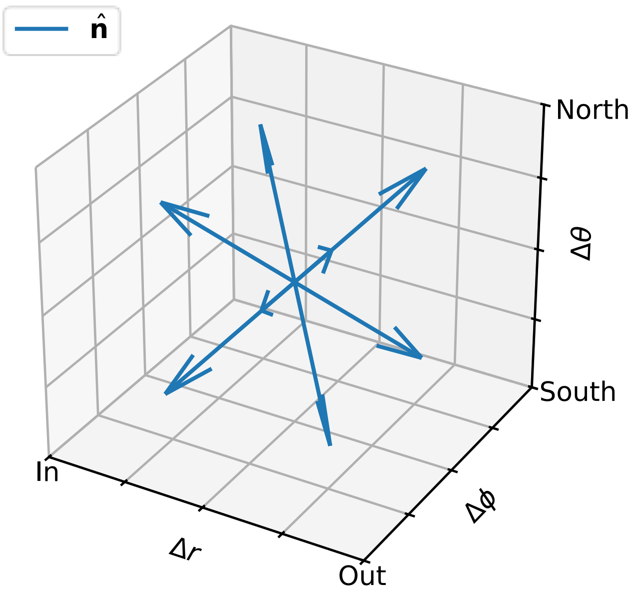
- No Hybrid
  - Ray-tracing now integrodifferential
- Multigroup
  - Percent level diff.
  - $N_f = 64$
  - $N_\zeta = 4$

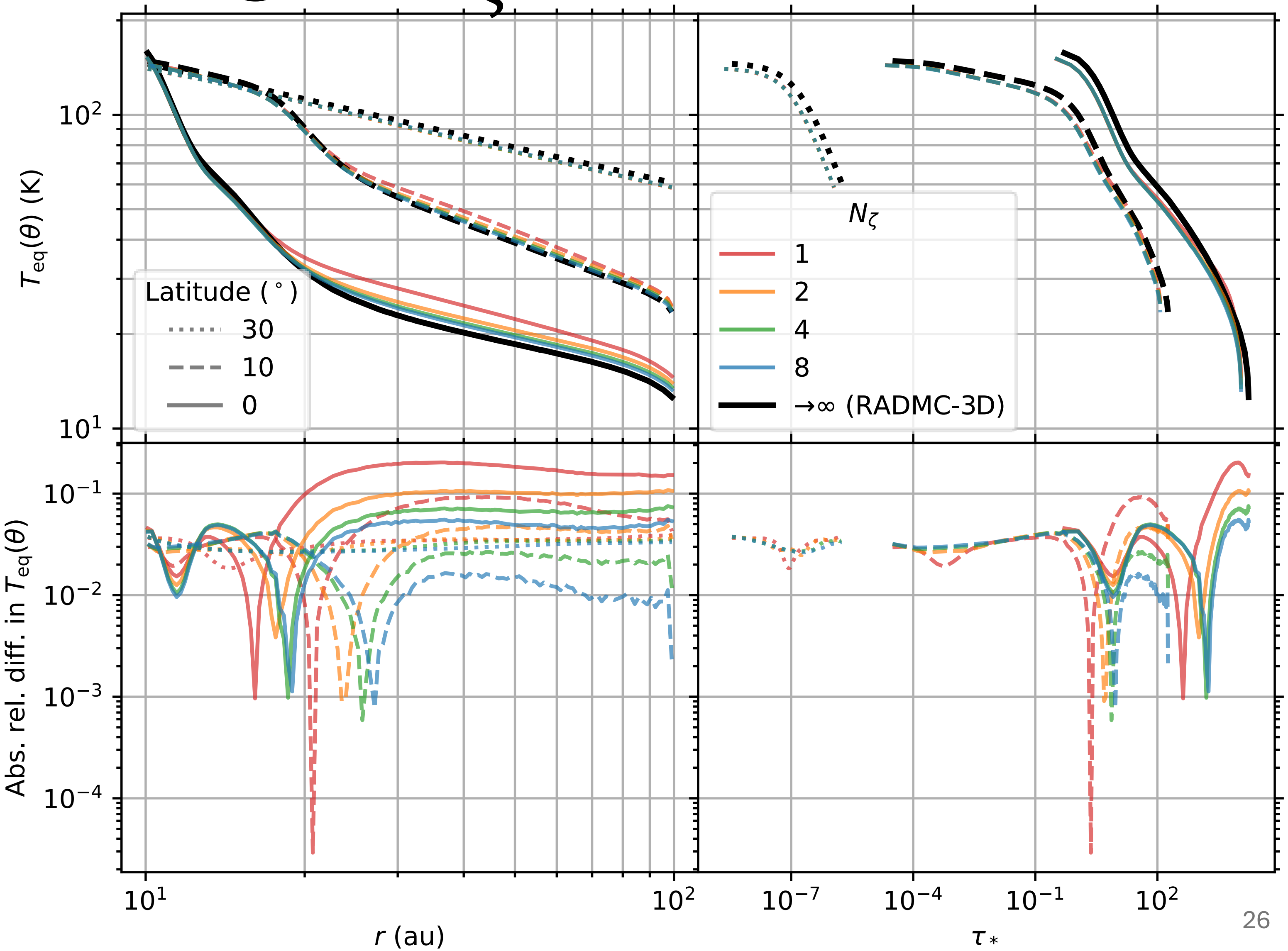


# Resolution and Performance

# Number of Polar Angles $N_\zeta$

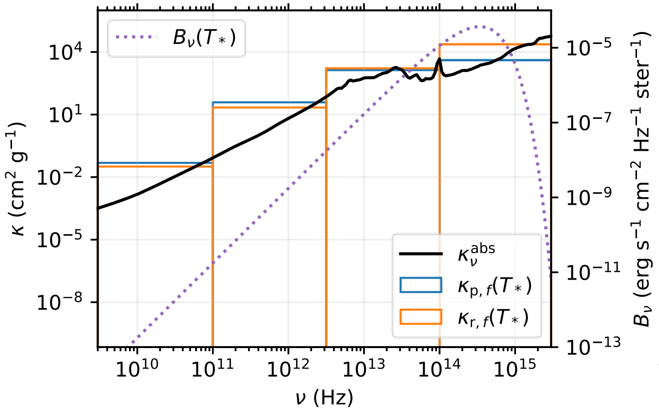
- Converged at low optical depth
- Closer to isotropic (re)emission at high optical depth



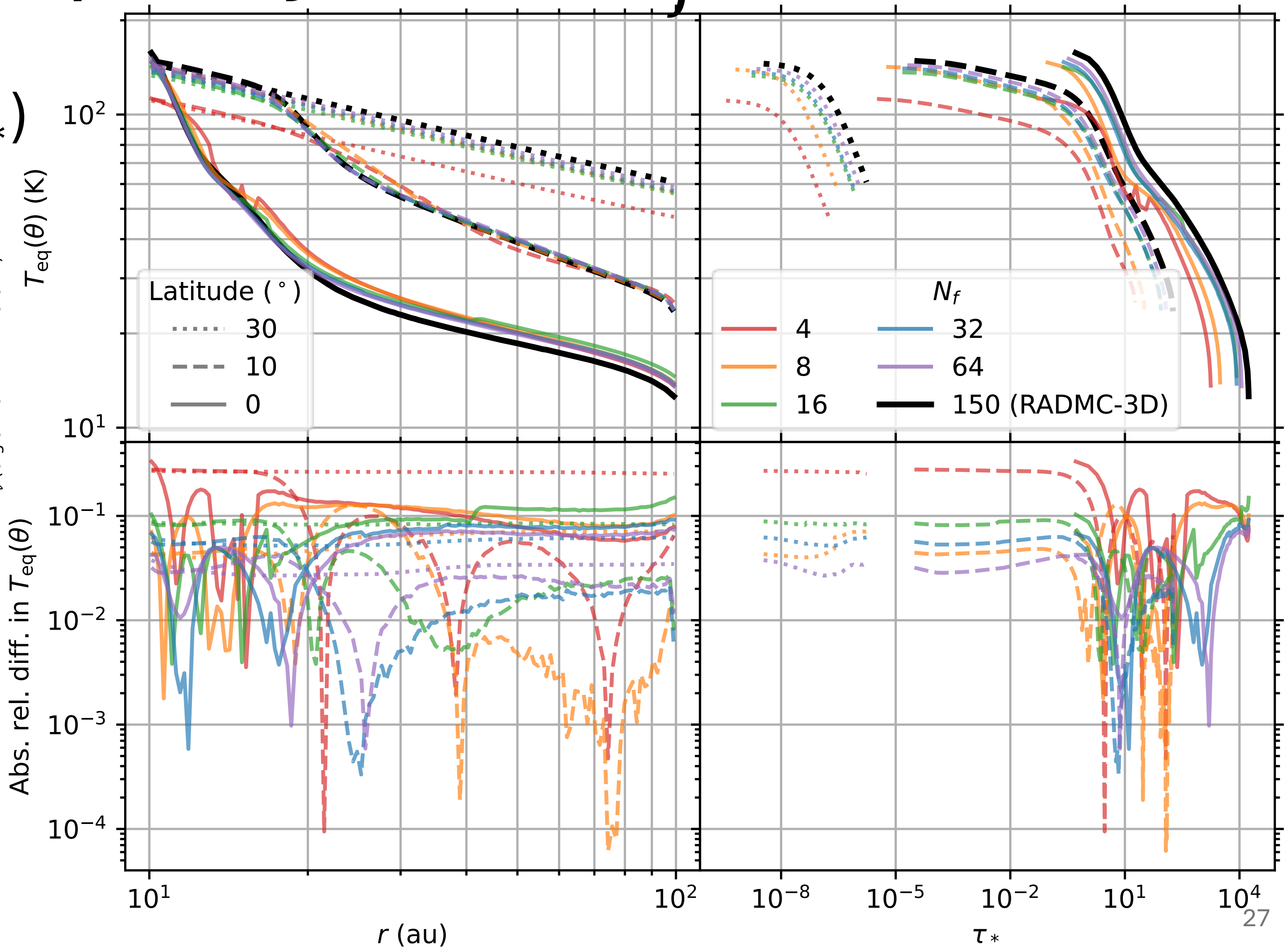



# Number of Frequency Bands $N_f$

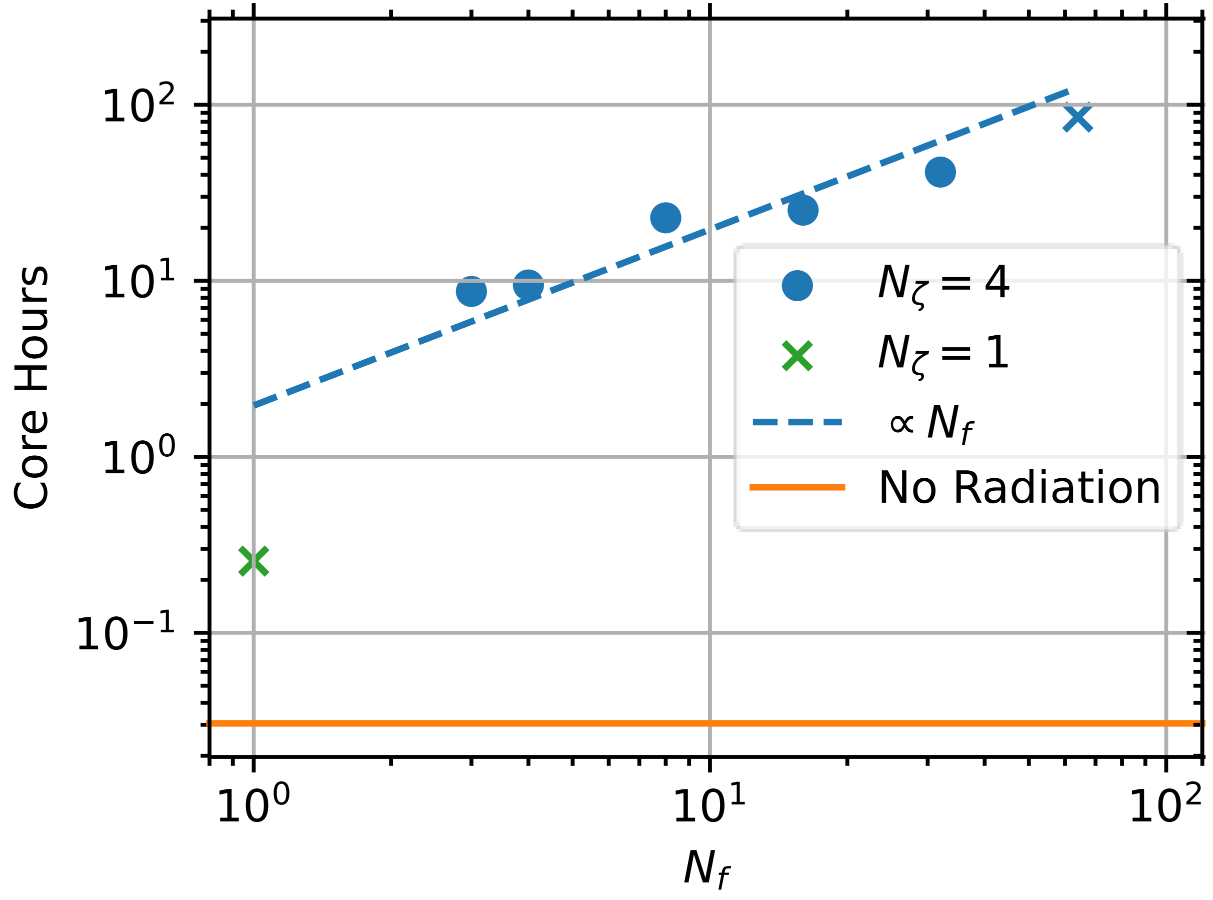
- Temp./rel. diffs. $(\tau_*)$
  - Not monotonic




  - $f_* \approx \nu_*$?
  - Band mean $\kappa_{f_*} \approx \kappa_{\nu_*}$?

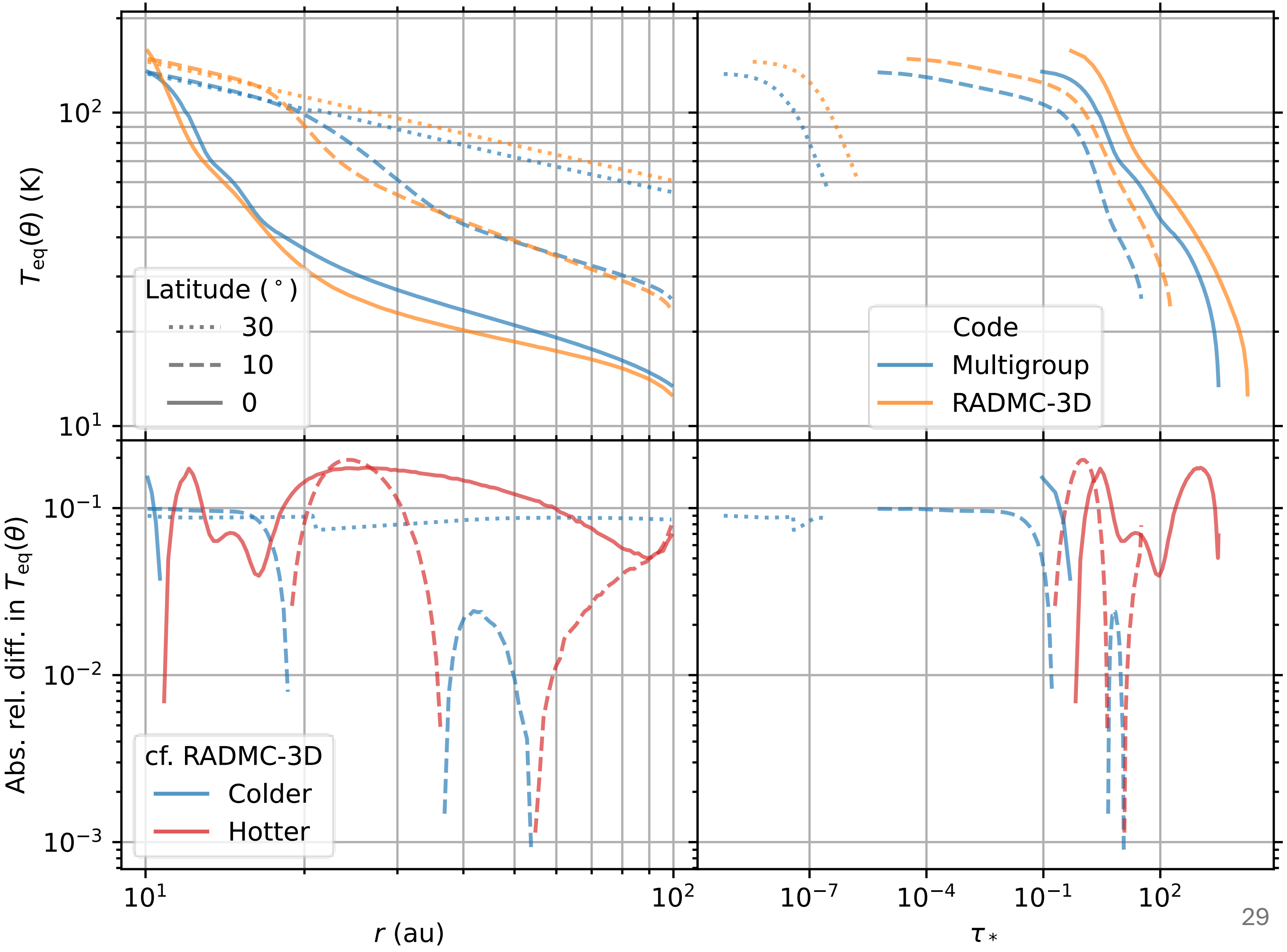


# Performance



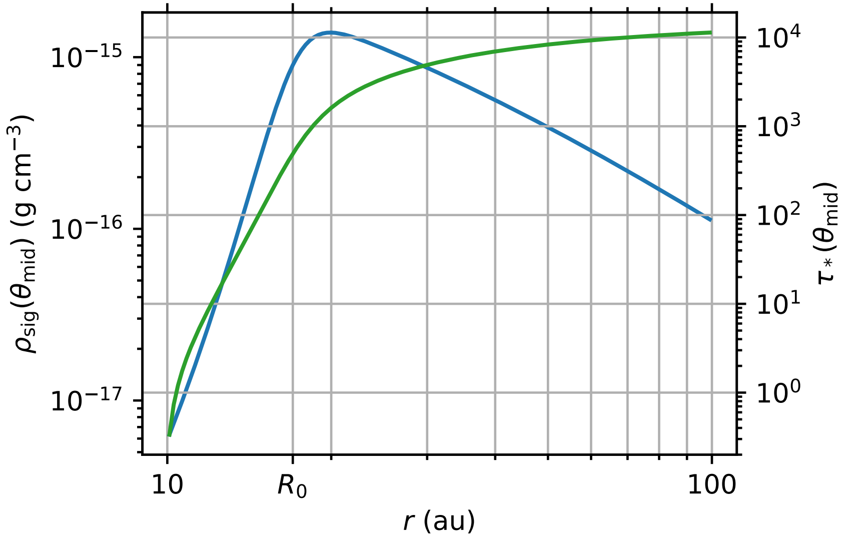
# $N_f = 3$ Accuracy vs. Performance

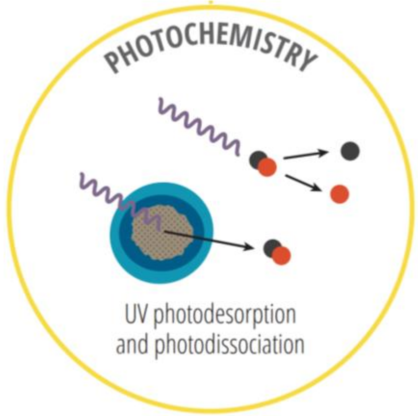
- ~2x less accurate
  - < 20% rel. diff.
- At least 10x fewer core hours



# Potential Applications

- Large-scale substructures
  - Inner dust cavity (current model)
    - Planet-carved gap(s)
    - Transition disks
- Variable luminosity
  - Accretion outbursts
    - Intense (sporadic?)
    - Frequency-dependent (Hα)
  - Circumbinary disks (periodic)
- Frequency-dependent disk chemistry



# Summary

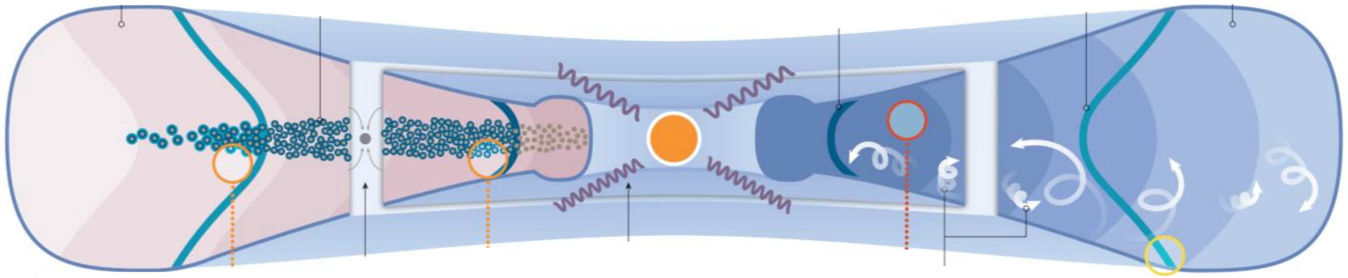
- **Athena++ framework:** multigroup rad. transport + radial stellar rays
- **Freq.-dependent opacity:** hotter atmosphere, cooler midplane
- **Scattering included:** >90% accurate across optical depths
- **Optimized 3-band model:** ≥10× cheaper, <20% diff.
- **Applications:** substructures, outbursts, chemistry, circumbinary

irrad_disk

arXiv:2606.08859

# Appendix

# Protoplanetary Disk Structure



Öberg+ 21

# Spherical Coordinates

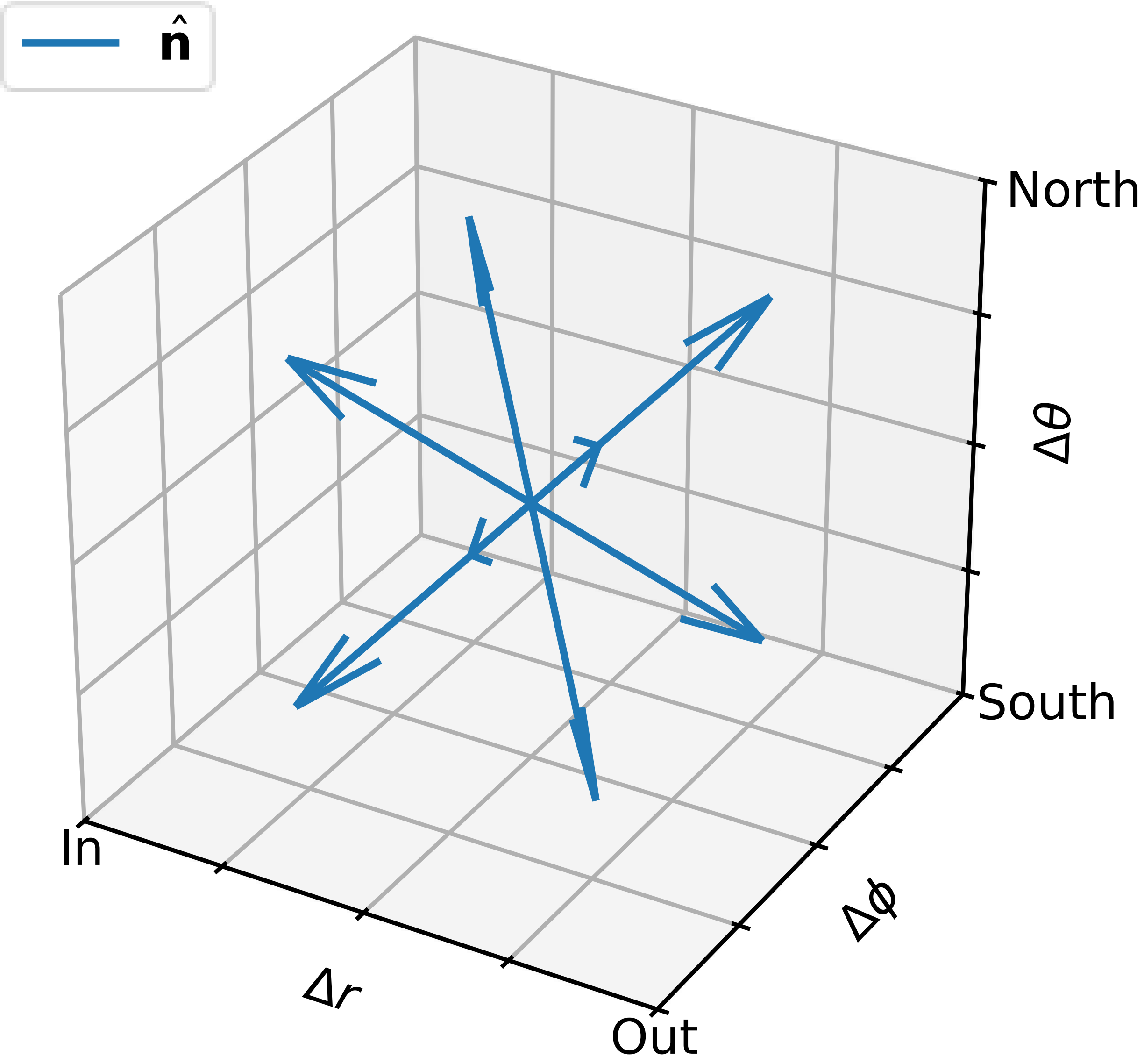
$N_\zeta \geq 1$ polar angles $(0,\ \pi/2)$

$N_\psi \geq 2$ longitudinal angles $(0,\ \pi)$

$N_{\text{ang}} = 4N_\zeta N_\psi$ angles per cell

$$\frac{1}{N_{\text{ang}}} \simeq \frac{\mathrm{d}\Omega}{4\pi}$$

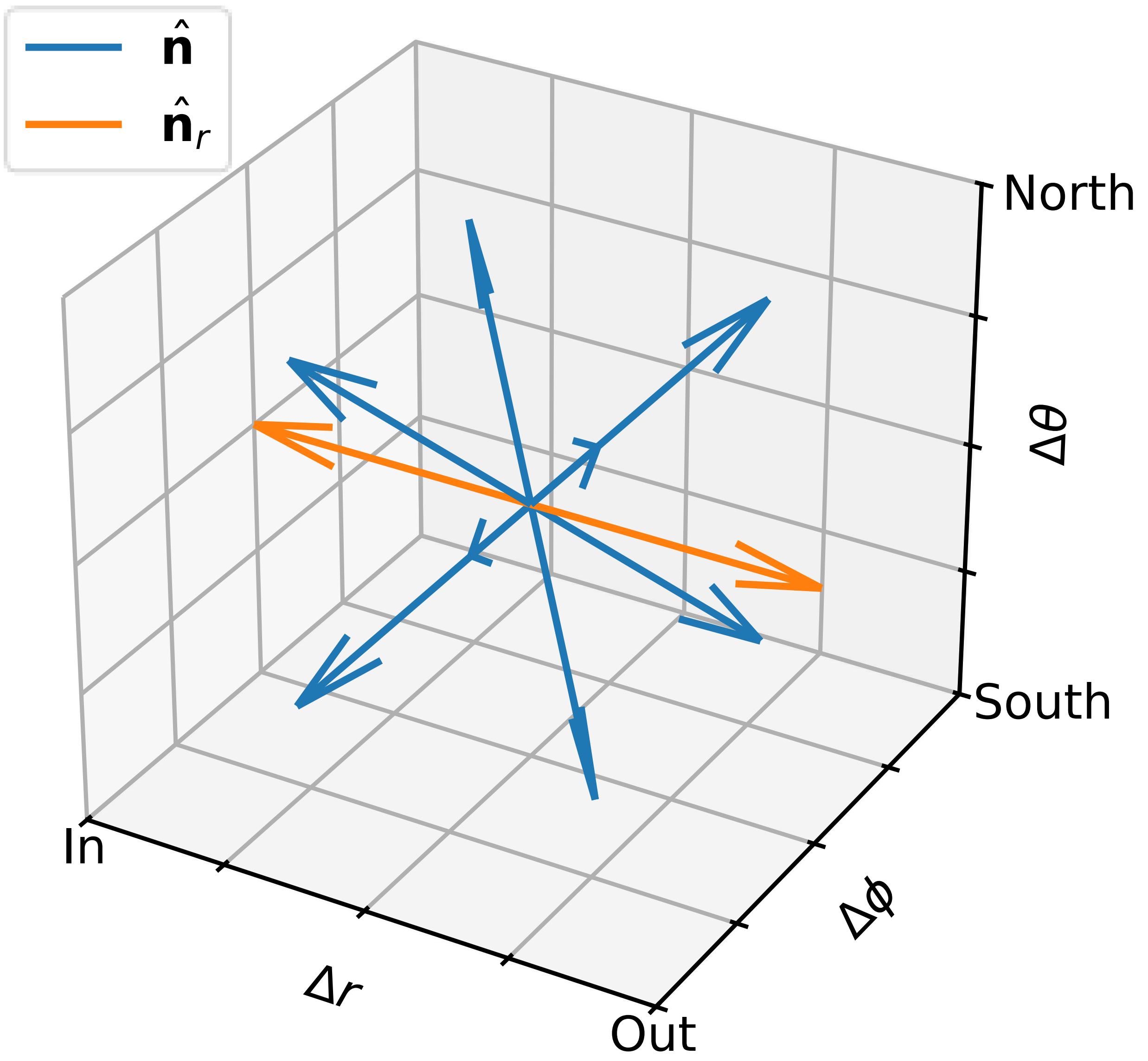
**Irradiative point source at origin?**



# Radial Rays

$N_{\text{ang}} = 4N_\zeta N_\psi + 2$ angles per cell

$$\frac{1}{N_{\text{ang}}} \simeq \frac{\mathrm{d}\Omega}{4\pi}$$



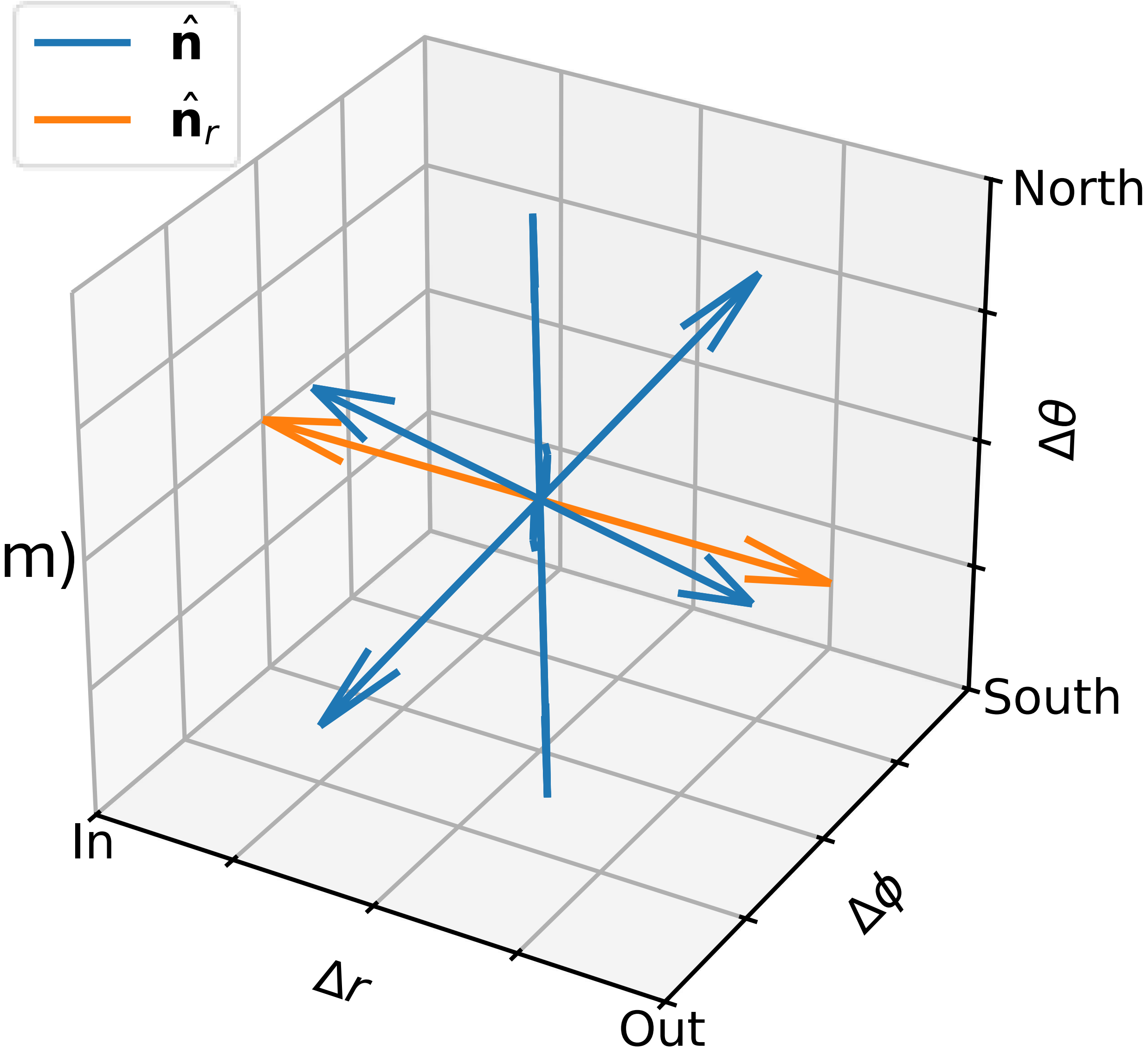
# Source Terms

**Lab frame**

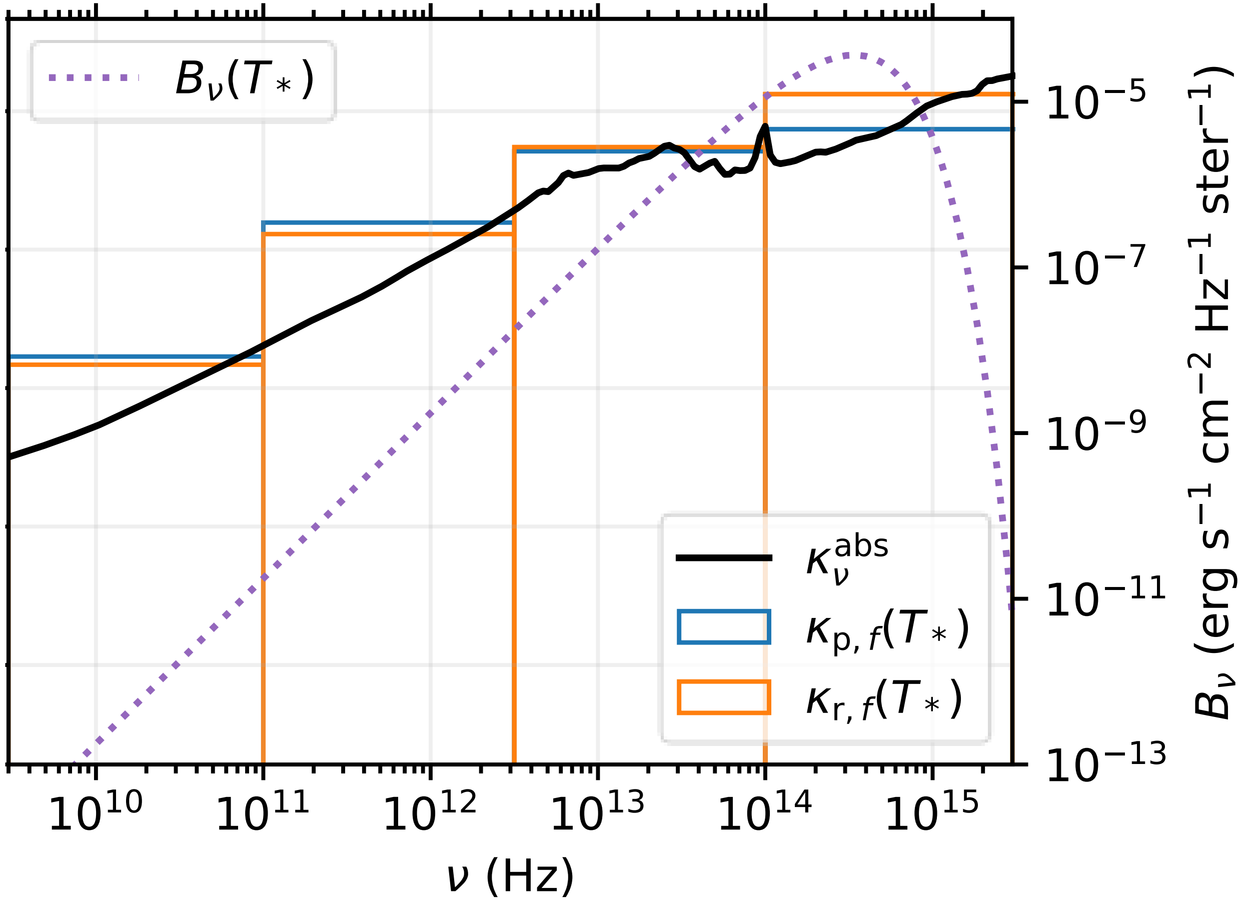
$$\frac{\partial I_f}{\partial t} + c\hat{\mathbf{n}} \cdot \nabla I_f = c\left(\overbrace{\eta_f}^{\text{emissivity}} - \overbrace{\chi_f I_f}^{\text{opacity}}\right)$$

**Comoving frame** (Lorentz transform)

$$\rho\left(\overbrace{\kappa_{\mathrm{s},f} + \kappa_{\mathrm{a},f}}^{\text{band-mean opacities}}\right)\left(J_{0,f} - I_{0,f}\right)$$
$$+ \rho\left(\kappa_{\mathrm{p},f}\, \underbrace{\varepsilon_{0,f}}_{\substack{\text{thermal}\\ \text{emissivity}}} - \kappa_{\mathrm{p},f,\mathrm{e}}\, \underbrace{J_{0,f}}_{\substack{\text{mean radiation}\\ \text{energy density}}}\right)$$



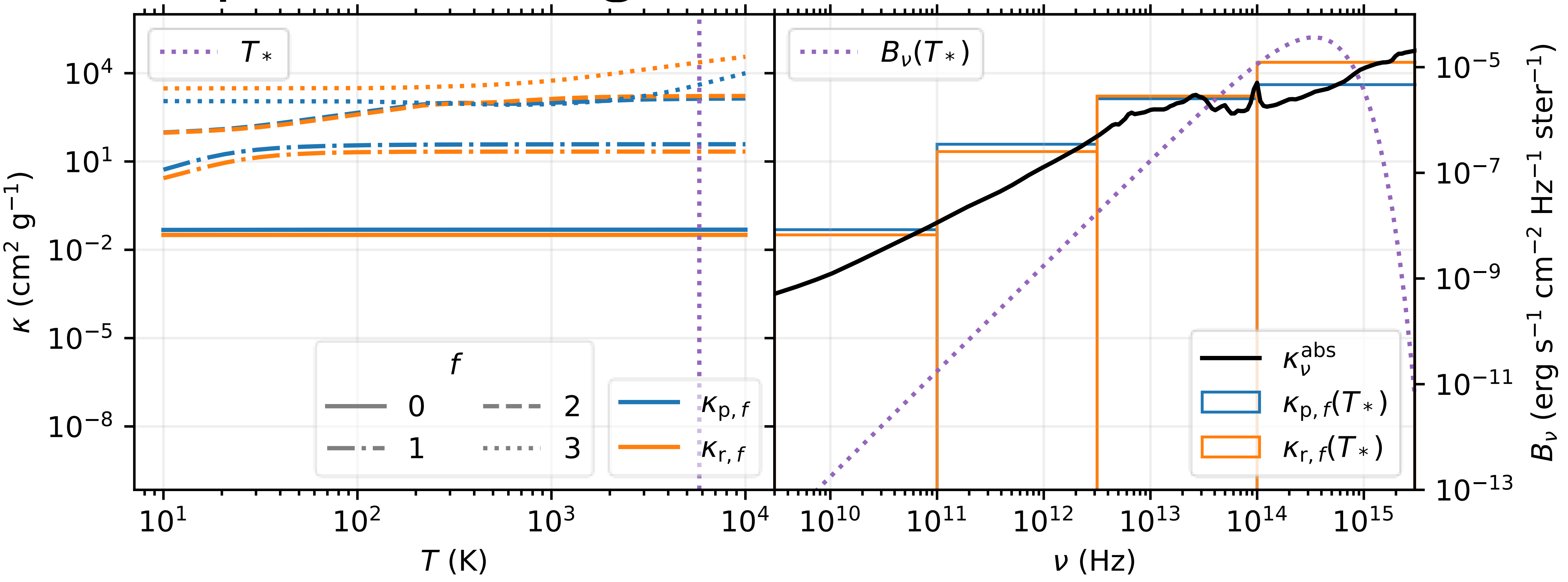
# Multigroup Frequency Bands ($N_f = 4$)




$$\kappa_{\mathrm{p},f}(T) \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \kappa_\nu^{\mathrm{abs}} B_\nu(T)\mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} B_\nu(T)\mathrm{d}\nu}$$

$$\frac{1}{\kappa_{\mathrm{r},f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\mathrm{abs}}\right)^{-1} (\partial B_\nu/\partial T)\mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu/\partial T)\mathrm{d}\nu}$$

# Temperature-weighted Band Means

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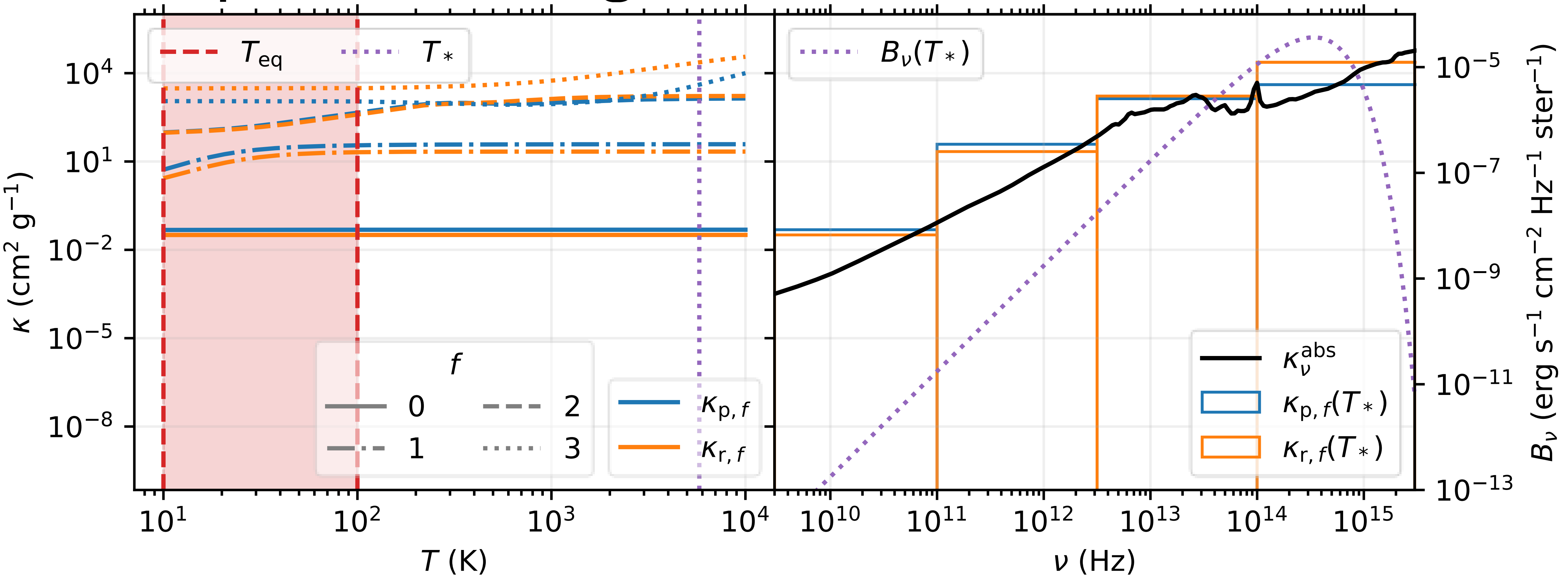


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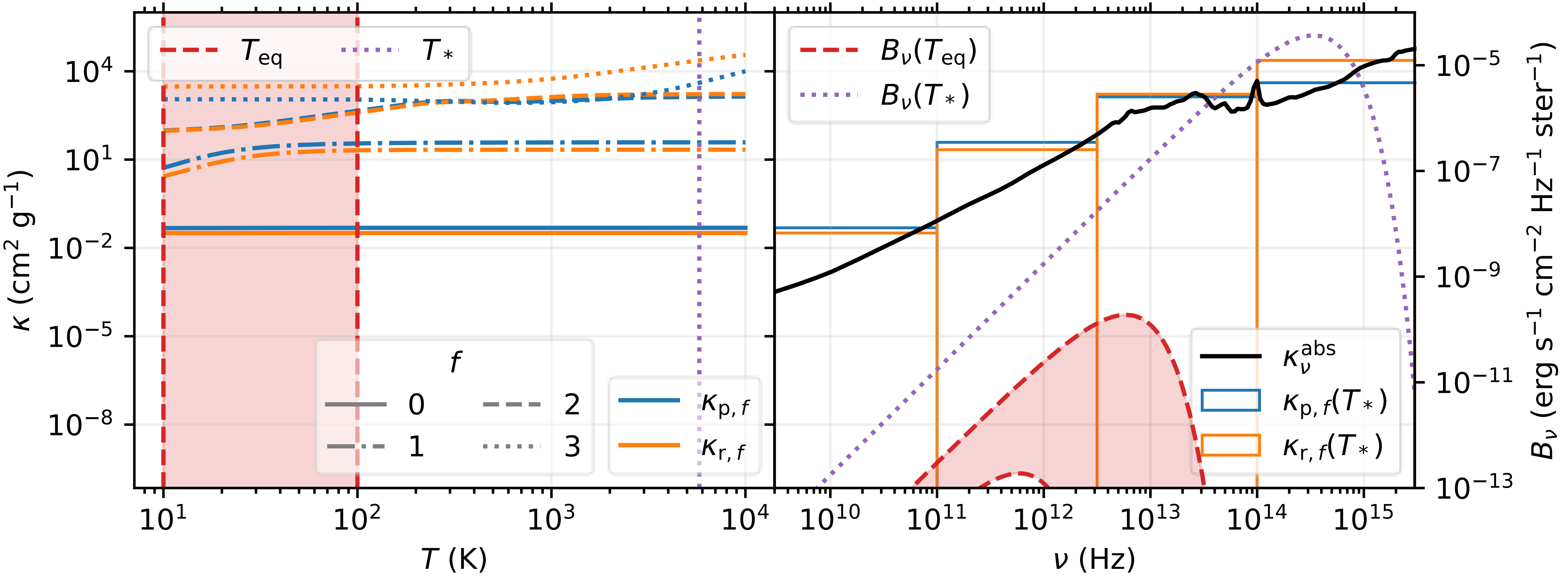
# Irradiative–Thermal Equilibrium Check

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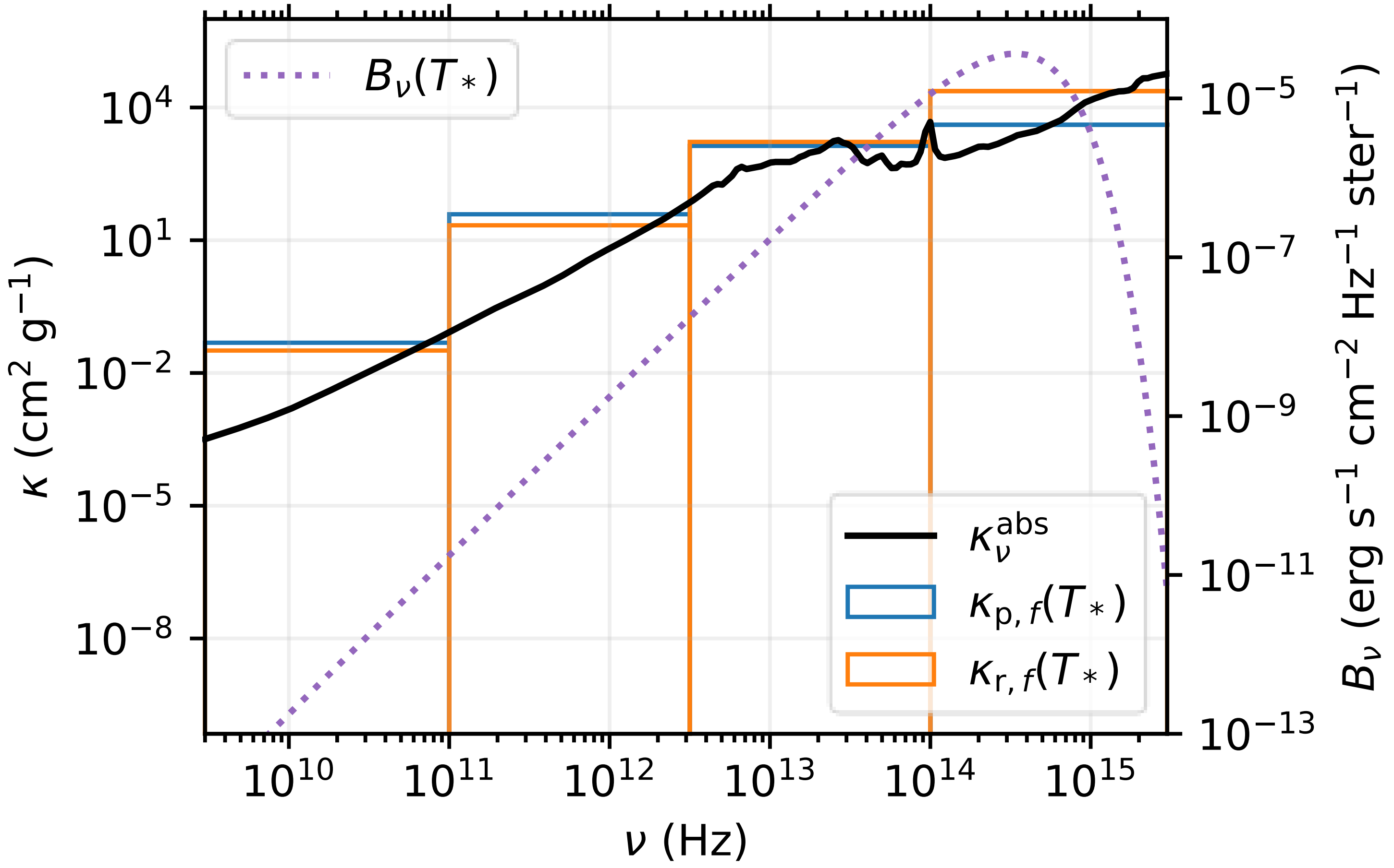
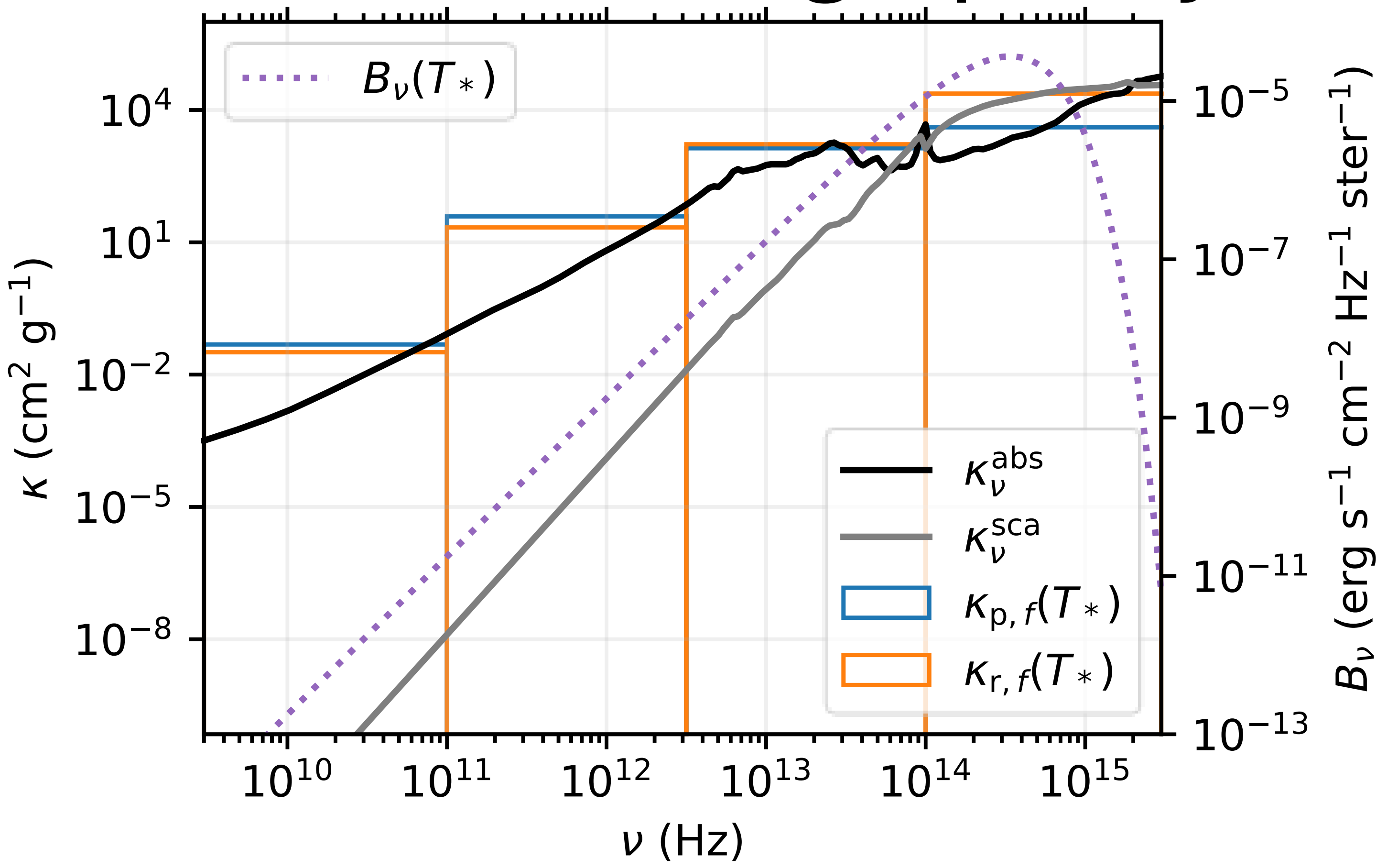
# Absorption Opacities

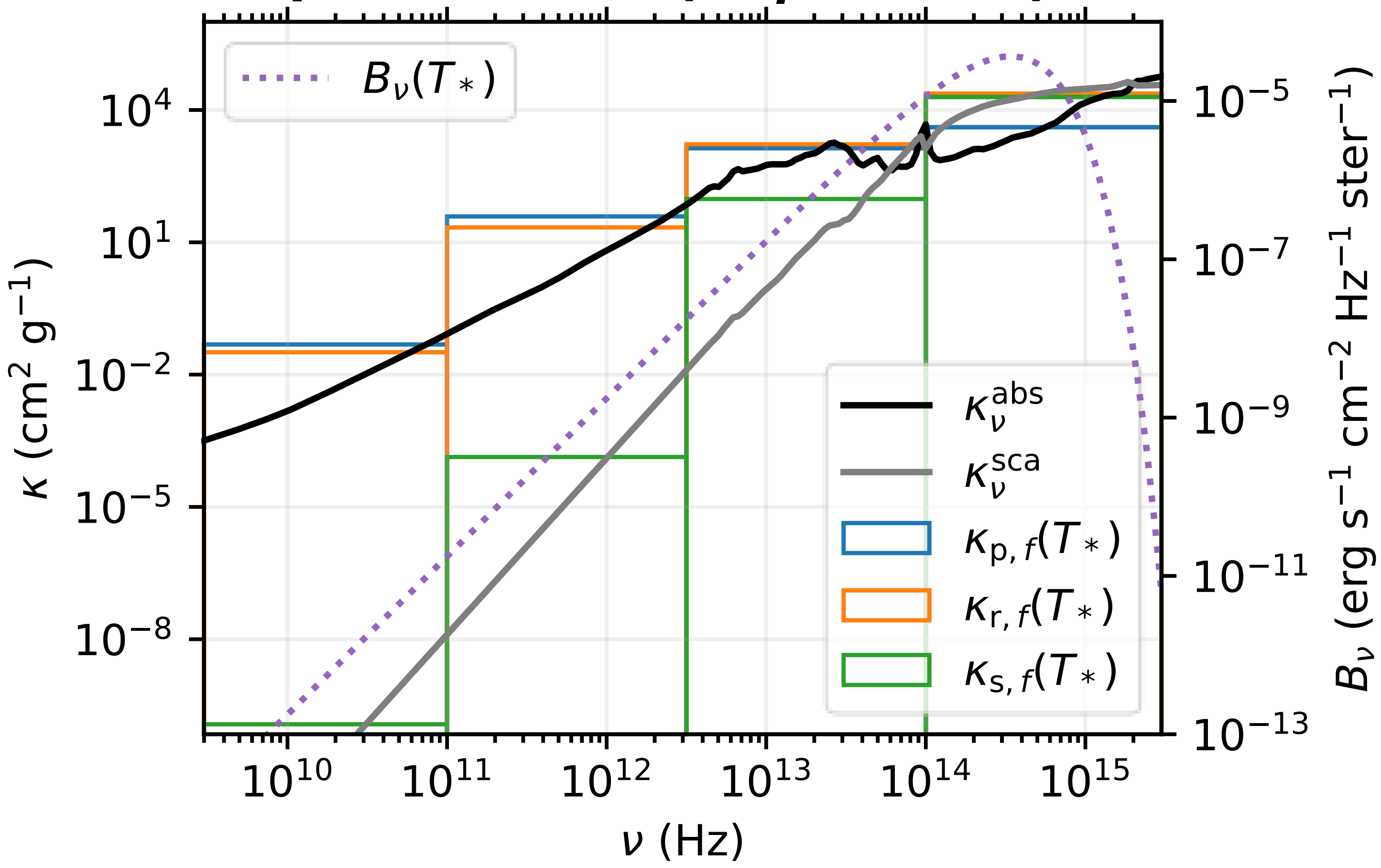


# Scattering

# Monochromatic Scattering Opacity



# Band-mean Opacities ($N_f = 4$)




$$\frac{1}{\kappa_{{\rm s},f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} (\kappa_\nu^{\rm sca})^{-1}\,(\partial B_\nu/\partial T)\mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu/\partial T)\mathrm{d}\nu}$$

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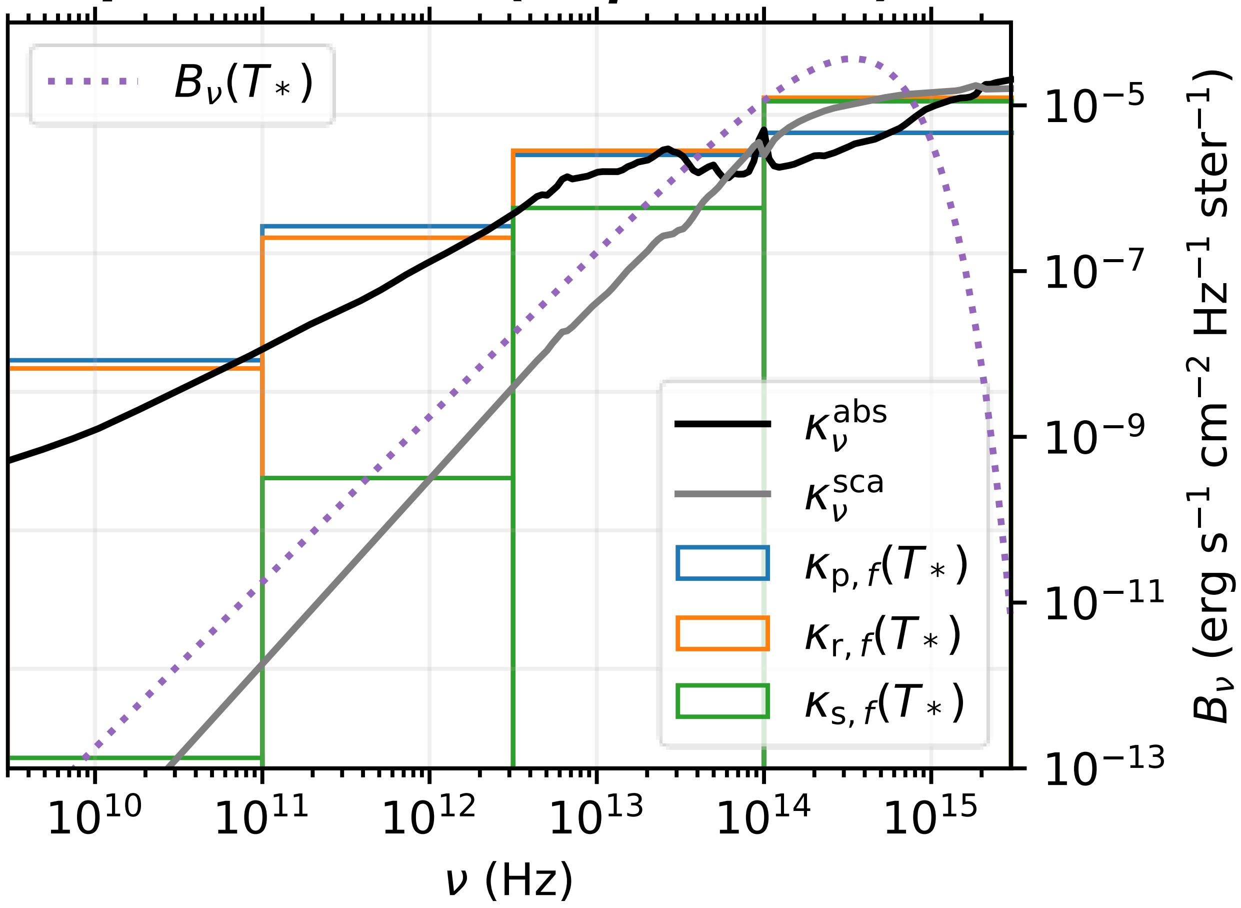


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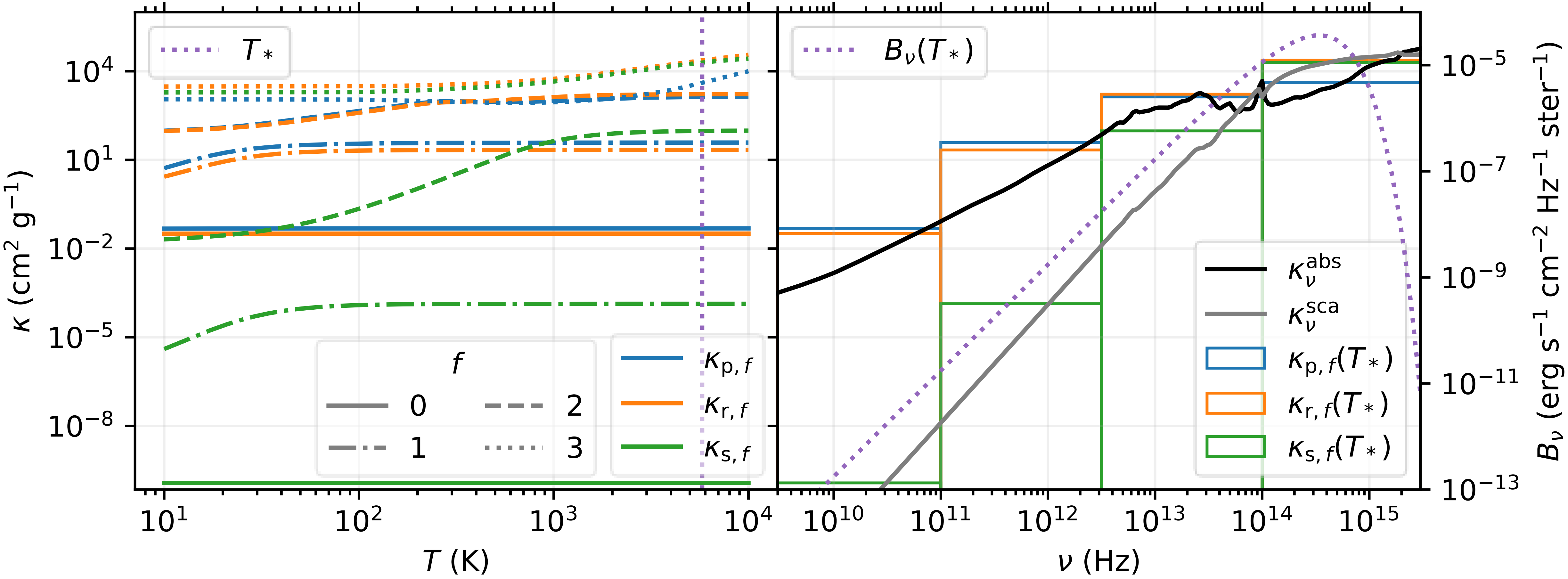
$$\frac{1}{\kappa_{r,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\text{abs}} + \kappa_\nu^{\text{sca}}\right)^{-1} \, (\partial B_\nu/\partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu/\partial T) \mathrm{d}\nu}$$

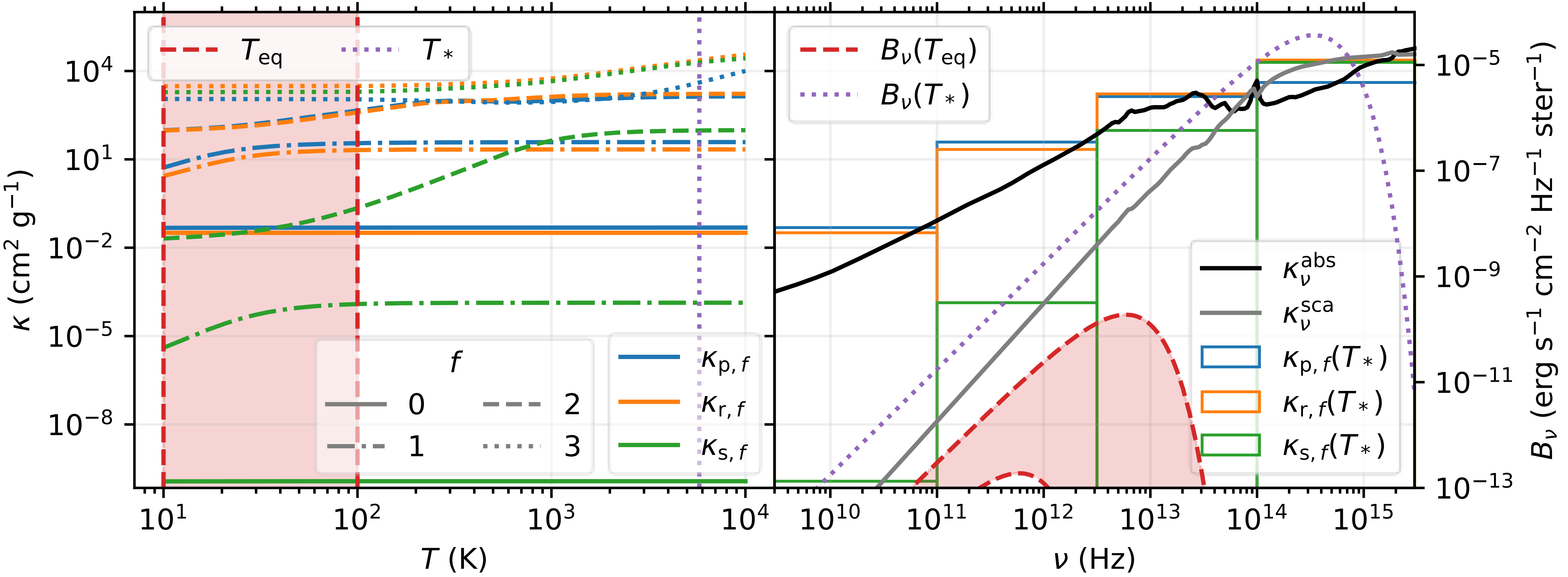
# Temperature-weighted Band Means




$$\frac{1}{\kappa_{s,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} (\kappa_\nu^{\mathrm{sca}})^{-1} (\partial B_\nu / \partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu / \partial T) \mathrm{d}\nu}$$

$$\frac{1}{\kappa_{r,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\mathrm{abs}} + \kappa_\nu^{\mathrm{sca}}\right)^{-1} (\partial B_\nu / \partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu / \partial T) \mathrm{d}\nu}$$

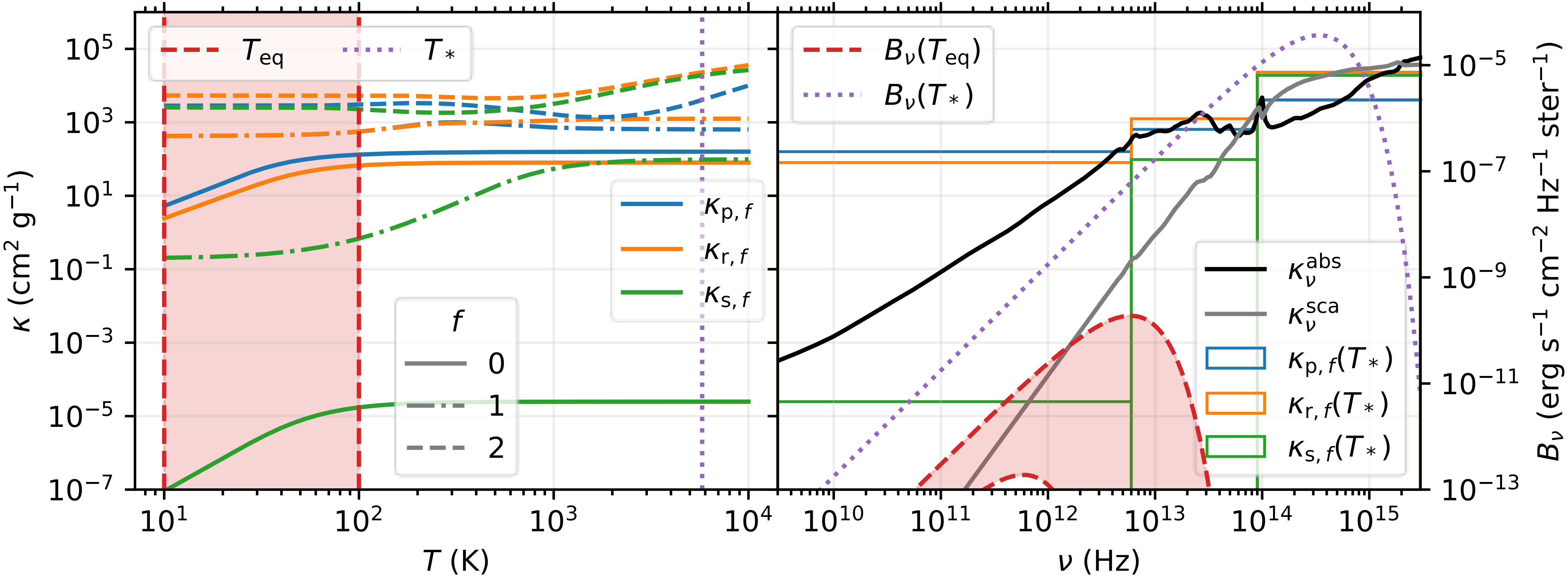
# Irradiative–Thermal Equilibrium Check




$$\frac{1}{\kappa_{s,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} (\kappa_\nu^{\rm sca})^{-1} \, (\partial B_\nu / \partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu / \partial T) \mathrm{d}\nu}$$

$$\frac{1}{\kappa_{r,f}(T)} \equiv \frac{\int_{\nu_f}^{\nu_{f+1}} \left(\kappa_\nu^{\rm abs} + \kappa_\nu^{\rm sca}\right)^{-1} \, (\partial B_\nu / \partial T) \mathrm{d}\nu}{\int_{\nu_f}^{\nu_{f+1}} (\partial B_\nu / \partial T) \mathrm{d}\nu}$$

# Optimized Band Mean Opacities ($N_f = 3$)



# Isotropic vs. Anisotropic Scattering

- Athena++ currently supports isotropic
- RADMC-3D only
  - Colder midplane
  - Warmer atmosphere
  - Anisotropic model
    - $g = \langle \cos(\theta) \rangle$ (Henyey & Greenstein 41)
  - < 10% rel. diff. w/ isotropic model